Pinhole Sunglasses: A Physically-Grounded Design and Evaluation Pipeline for Multi-Aperture Myopia Correction

IYPT 2027 – Problem 1 – Full Research Compilation

Prepared from the Pinhole Sunglasses project repository

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Abstract

A pinhole reduces the blur circle a defocused (myopic) eye forms on the retina by restricting the bundle of rays that reach any one retinal point – the stenopeic effect – without requiring a corrective lens. This project builds, validates, and experimentally tests a complete physical pipeline for pinhole-based myopia correction, from a single aperture to a full multi-aperture array intended to restore a normal field of view. Single-aperture blur is modelled first by a closed-form root-sum-of-squares (RSS) approximation, then superseded by an exact Angular Spectrum Method (ASM) wave-optics simulation after the RSS model was measured to diverge from the exact result by 64.8% on average (up to 192.8%) and a specific implementation defect – a single Fourier transform silently evaluated at the eye's true focal plane rather than the myopic retina – was found and corrected. Predicted blur is converted to predicted visual acuity (LogMAR) via a cited contrast-sensitivity and brightness model. Two independent array-safety evaluation frameworks are developed: Model A, a single-axis worst-case gate; and Model B, a full two-dimensional, empirically gaze-time-weighted evaluation that generalizes Model A's single-axis assumption and becomes this project's primary standard following an explicit, documented methodology decision. Both models are applied to eleven array geometry families; hexagonal packing emerges as the safest choice for six of eight tested myopia powers, and this result is connected to a first-principles argument from lattice symmetry (the crystallographic restriction theorem) and demonstrated with matched-spacing real-geometry data. A comprehensive ghosting theory (Dirac-comb array spectrum, Barten (1999) contrast sensitivity) reverses an earlier heuristic-CSF-based safety classification for the entire tested dense-pitch range, and a harmonic-series extension shows the fundamental ghosting frequency is never the most dangerous one. Single-aperture predictions are validated against original photographic and optical-bench experiments; array-level predictions are validated against a real, though small and non-per-subject-resolved, published clinical dataset (Wang, 2009), against which the model's raw form fails a formal $R^2 > 0$ significance bar at every tested condition, and a case study following one specific bug (an optics import-shadowing defect) is used to show that model accuracy against that same external dataset improved measurably, though not sufficiently, once corrected. This paper documents the pipeline, every quantitative result with its file-level provenance, four independently-identified and corrected implementation defects with full before/after evidence, and an honest account of what remains unresolved.

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Chapter 1

Introduction and Problem Statement

1.1 The official problem

IYPT 2027 Problem 1 ("Pinhole Sunglasses") asks: myopic (short-sighted) people cannot focus light from distant objects onto the retina without a corrective lens; investigate whether an array of small apertures – a pinhole occluder built into a wearable frame – can substitute for a corrective lens over a useful field of view, and characterize the trade-offs involved.[1] A single small aperture is well known to improve the perceived sharpness of a defocused image (the pinhole or stenopeic effect, Section 2.2); the open engineering question this project addresses is what happens when that single aperture is replaced by an *array* of apertures large enough, in aggregate, to give a normal field of view, while each individual aperture remains small enough to retain the stenopeic benefit.

1.2 Why this is genuinely hard

A single pinhole trades brightness and field of view for depth of focus. An array of pinholes must additionally avoid two new, array-specific failure modes that do not exist for a single aperture at all:

- **Ghosting.** If the eye's pupil simultaneously admits light through two or more apertures, it forms two or more laterally-shifted copies of the same retinal image – double or multiple vision.
- **Blinding.** If the eye's pupil, at some gaze direction, admits light through *zero* apertures, no image reaches the retina at all at that gaze direction – a genuine dark gap in the field of view.
- **Vignetting.** As the eye rotates to look through different parts of the array, the pupil physically shifts relative to the (head-fixed) aperture pattern, changing which apertures are active – the wearer's own eye movements are themselves part of the optical system, not a fixed viewing condition.

None of these three effects has a single-aperture analogue; all three are properties of the *arrangement* of apertures, not of any one aperture's own optics. This project treats single-aperture optics (Chapters 2–4) and array-level geometry (Chapters 5–8) as two genuinely separate physics problems, connected only by the fact that every aperture in a real array shares identical per-hole optics with a standalone single aperture of the same diameter – a claim used repeatedly below and verified directly in Section 2.11.

[1] Problem statement summarized from `IYPT_Pinhole_Sunglasses_Problem_Statement.md` (repository root, verified present; see Section 13 for full provenance).

1.3 Design variables and constraints

The design variables available to this project are: the aperture diameter a_{hole} (per myopia power M, since the optimal aperture is itself a function of the wearer's own refractive error); the array *family* (the geometric rule generating hole positions – square, hexagonal, ring-based, spiral-based, and others, Section 5.1); the array pitch or spacing; and, where relevant, the lattice rotation. The physical constraints are the wearer's own pupil diameter (taken as $D_{\text{pupil}} = 4000\,\mu\text{m}$ throughout, a standard photopic value), the reduced-eye power $P_0 = 60\text{D}$, and the gaze range over which the design must remain safe (this project evaluates $\varphi \in [0°, 25°]$ eccentricity, with one bin at 27° used and explicitly flagged as outside the validated small-angle range of the underlying pupil-shift formula – Section 5.5).

1.4 What "success" means for this document

Every number in this paper is required to trace to a specific file in the project repository, cited by path and by that file's modification time, so that a reader can independently confirm it is the current, not a superseded, result (Section 0 of the task brief that produced this document; see the Data Provenance Appendix, Chapter 14). Four real implementation defects were found and corrected during the project's own development; rather than silently updating results and moving on, this paper documents each one as a worked case study (Chapter 10), because the corrections are themselves part of the project's contribution – they show where a plausible-looking model was wrong, why, and by how much.

Chapter 2

Theoretical Framework I: Single-Aperture Optics

2.1 The reduced eye model

Every optical calculation in this project uses the standard *reduced eye* simplification: the cornea, aqueous, crystalline lens, and vitreous are lumped into a single refracting surface of power P_0, situated at one plane, imaging directly onto the retina. This is the same simplification underlying the Gullstrand–Emsley reduced eye and is what makes a single scalar P_0 meaningful at all; it deliberately does not model the cornea/lens power split (roughly 2:1 in a real eye), nor the physical anterior-chamber depth ($\approx 3.6\,\text{mm}$) separating them. This choice is stated explicitly in `stage2_fixed/phase2_retina.py`'s own module docstring (PHYSICS_AUDIT.md item E1, verified present in that file, mtime – see Chapter 14) precisely so a reader cannot mistake it for an oversight.

The values used throughout are:

Symbol	Value	Meaning
P_0	60.0 D	Total (emmetropic) reduced-eye power
n'	1.336	Refractive index of the vitreous humour
D_{pupil}	4000 μm	Photopic pupil diameter
λ	550 nm	Vacuum design wavelength (photopic luminous peak)
d_{gap}	12 mm	Spectacle-plane vertex distance, pinhole to cornea

(source: `stage2_fixed/myopia_axial.py`, `new_sweeps/run_acuity_sweep.py` constants `WAVELENGTH_NM=550.0`, `D_GAP_MM=12.0`; see Chapter 14 for exact mtimes).

2.1.1 Axial myopia model

This project models myopia as *axial*, not *refractive*: the eye's total power P_0 is held fixed at its emmetropic value for every degree of myopia M; only the eye's axial length grows, pushing the retina past the eye's own (unchanged) focal point. This is the defining choice of `stage2_fixed/myopia_axial.py` (`axial_length_for_myopia`, verified read directly):

$$L_0 = \frac{n'}{P_0} \qquad \text{(emmetropic axial length, fixed reference)} \tag{2.1}$$

$$L_{\text{ax}}(M) = \frac{n'}{P_0 - M} \qquad \text{(myopic axial length for this } M\text{)} \tag{2.2}$$

$$f_{\text{eye}} = L_0 \qquad \text{(eye power fixed at the emmetropic value for every } M\text{)} \tag{2.3}$$

The alternative – a *refractive* model, in which P_0 itself changes with M while axial length stays fixed – was not implemented; the axial convention was chosen because it is the anatomically dominant mechanism in real myopia (axial elongation of the globe) and because it lets one physical eye power P_0 be reused unmodified across the entire M sweep. With $P_0 = 60\text{D}$ and $n' = 1.336$: $f'_{\text{image}} = n'/P_0 \times 10^6 = 22{,}266.667\,\mu\text{m}$ – the media-reduced image distance, not the naive vacuum value $1/P_0$.

2.2 Geometric blur, diffraction blur, and the stenopeic effect

An uncorrected myopic eye focuses light from a distant object *in front of* the retina; by the time the diverging cone reaches the actual retinal plane it has spread into a geometric blur circle of radius approximately proportional to the eye's pupil diameter and to the degree of defocus. A pinhole of diameter $a_{\text{hole}} \ll D_{\text{pupil}}$ placed in front of the eye restricts the admitted ray bundle to a much narrower cone, shrinking that geometric blur circle in direct proportion to a_{hole} – the stenopeic ("narrow-aperture") effect, and the entire physical basis for treating a small aperture as a myopia correction at all.

This benefit is not free. Diffraction at the aperture itself spreads the beam by an angle inversely proportional to a_{hole}: shrinking the aperture to suppress geometric blur simultaneously grows diffraction blur. A correctly-optimized aperture therefore sits at neither extreme, but at whichever size minimizes the two effects' combination for a given myopia power M – the entire quantitative content of Sections 2.3–2.4 below, and the reason "smaller is not simply better" (title of the corresponding `MASTER_HANDBOOK.tex` section, Chapter 2, verified present).

2.3 The RSS combination model and its limits

The first, closed-form model this project implemented combines the two blur mechanisms as an independent-error root-sum-of-squares (RSS):

$$c_{\text{geom}} = \frac{a_{\text{hole}}\, M}{P_0} \tag{2.4}$$

$$c_{\text{diff}} = 1.22\, \frac{\lambda_{\text{eff}}\, f'}{a_{\text{hole}}} \tag{2.5}$$

$$c_{\text{RSS}} = \sqrt{c_{\text{geom}}^2 + c_{\text{diff}}^2} \tag{2.6}$$

with $\lambda_{\text{eff}} = \lambda/n'$ the media-reduced wavelength and f' the media-reduced eye focal length. This form is attributed in this project's own code and documentation to Shen (2026) [Shen, 2026], whose paper describes it explicitly as an *engineering approximation*, not an exact result – language this project takes at face value rather than treating the formula as exact.

Measured divergence from the exact result

Directly measured against the exact ASM propagation (Section 2.4) across the project's own sweep grid: **64.8% mean divergence, up to 192.8% maximum**, worst specifically at *mild* myopia (source: `Array_Design/array_geometry_lab/Physics_Guide/MASTER_HANDBOOK.tex`, Chapter 6, "Deprecated formulas, and the error each one carries" table; verified present, see Chapter 14). The RSS model is retained in this project's codebase for historical comparison and cross-checking against Shen's own published numbers (Chapter 7, hexagonal-packing discussion), but is **deprecated as a predictive blur model** in favour of the exact method below.

The RSS model's underlying assumption – that geometric and diffraction blur are two statistically independent Gaussian error sources whose variances simply add – is false on its face for this problem: neither blur mechanism here is Gaussian (geometric blur from a circular aperture is a hard-edged disc; diffraction blur is an Airy pattern), and the two mechanisms are not statistically independent draws but two deterministic transformations of the *same* coherent field. RSS combination is a convenient closed form, not a physical derivation.

2.4 The Angular Spectrum Method: exact propagation

The correct approach solves the scalar Helmholtz equation exactly between parallel planes, with no near-field or far-field approximation. Any field $U_0(x, y)$ at a plane decomposes into plane-wave (angular-spectrum) components, each acquiring a phase set by its own transverse spatial frequency as it propagates a distance z:

$$U(x, y, z) = \mathcal{F}^{-1}\Big\{ \mathcal{F}\{U_0(x, y)\} \cdot H(f_x, f_y; z) \Big\}, \qquad H(f_x, f_y; z) = \exp\Big(ikz\sqrt{1 - \lambda^2(f_x^2 + f_y^2)}\Big) \tag{2.7}$$

valid for the propagating band $f_x^2 + f_y^2 \leq 1/\lambda^2$; beyond that the square root becomes imaginary and the corresponding (evanescent) components are dropped ($H = 0$), since they decay to negligible amplitude over any macroscopic propagation distance in this problem. Equation 2.7 is exact – it is the full solution of the free-space Helmholtz equation subject only to the scalar (non-vector) field approximation, which is well justified here since every aperture in this project is many wavelengths across.

This project uses Equation 2.7 for *both* propagation stages in the optical chain (mask plane to cornea; eye lens to retina), rather than a single closed-form approximation, because the two stages sit in genuinely different diffraction regimes (Section 2.5) and no single closed form serves both correctly.

2.4.1 The pipeline chain

The full per-point chain, exactly as implemented in `stage2_fixed/phase2_retina.py`'s `run_pipeline_point` (verified read directly, function signature and body):

1. **Mask to cornea**: a circular aperture is ASM-propagated across the spectacle gap d_{gap} to the corneal plane (`run_gap_to_cornea`).
2. **Pupil clip**: the field is truncated to the (larger) physiological pupil diameter D_{pupil} (`pupil_clip`) – a binary, hard-edged mask (no Stiles–Crawford directional apodization at this stage; see Section 9).
3. **Lens phase**: a quadratic phase screen corresponding to the eye's fixed power P_0 is applied (`lens_phase`) – pure defocus/quadratic phase only, no higher-order aberration terms.
4. **Lens to retina**: the field is ASM-propagated the myopic axial distance $v_r = L_{\text{ax}}(M)$ to the (now-defocused) retinal plane (`propagate_to_retina`).
5. **Retinal intensity**: $I_{\text{retina}} = |U_{\text{retina}}|^2$, from which the encircled-energy blur radius and, downstream, the visual acuity prediction (Chapter 3) are computed.

2.4.2 The effective-wavelength correction

The lens phase screen and the retina-propagation step must use the *same* effective wavelength – $\lambda_{\text{eff}} = \lambda/n'$ when `refractive_index_correction=True` (this project's default), else the raw vacuum wavelength λ. This effective wavelength is resolved *once*, in `run_pipeline_point`, and

threaded through both the lens-phase and retina-propagation calls – a design choice made specifically because an earlier version of this project's code did not do this (Section 10.1, the single-FFT defocus-plane bug and its root cause both trace to exactly this wavelength-consistency issue).

2.5 Fresnel number: which diffraction regime a given stage is in

$$N_F = \frac{R_{\text{hole}}^2}{\lambda z} \tag{2.8}$$

$N_F \ll 1$ corresponds to the far field (a clean Airy pattern); $N_F \sim$ 1–10 is the Fresnel transition regime (several comparably-bright diffraction rings); $N_F \gg 10$ is near-geometric (a sharp-edged projected patch, diffraction a small correction). This project's own sweeps span Fresnel numbers well under 1 to over 20 across the full aperture×myopia grid (source: `Array_Design/array_geometry_lab/Physics_Guide/MASTER_HANDBOOK.tex` Chapter 6) – i.e. the mask-to-cornea and lens-to-retina propagation stages routinely sit in *different* regimes from each other within a single evaluated point, which is the direct justification for using the regime-agnostic exact method (Equation 2.7) rather than a regime-specific closed form for either stage.

2.6 Numerical implementation: grid, band-limiting, and energy accounting

The ASM integral is evaluated on a finite square grid of $N \times N$ samples spanning a physical window L (so pixel pitch $dx = L/N$), via FFT. Two numerical issues require explicit handling, both already present in the implementation and both logged, not silently corrected:

Band-limiting. A finite grid cannot represent spatial frequencies beyond its own Nyquist limit; a hard-edged pinhole mask has high-frequency content that, left unfiltered, aliases into spurious fine structure that is numerically indistinguishable from genuine diffraction structure. The Matsushima band limit [Matsushima and Shimobaba, 2009] suppresses spectral content beyond the angle at which this aliasing would occur; every run logs the suppressed energy fraction explicitly, rather than treating it as an unaccounted loss.

Why the energy-conservation check itself needed correcting

Since the propagation kernel H in Equation 2.7 has unit modulus on every retained (propagating) spectral component and is exactly zero on every suppressed (evanescent or band-limited) one, the apparent energy loss after propagation is *identically* the band-limit-suppressed spectral fraction: $\varepsilon_{\text{energy}} \equiv f_{\text{suppressed}}$ – confirmed against this project's own sweep data at correlation 1.000000 (source: `MASTER_HANDBOOK.tex` Chapter 6, "Numerical band-limiting"). An energy-conservation test written against a flat tolerance ("energy in must equal energy out to within ϵ") would therefore fail on *every single row* from this intentional, already-accounted-for suppression alone, never distinguishing it from a genuine bug. The corrected test instead checks the *residual* $\varepsilon_{\text{energy}} - f_{\text{suppressed}}$, which is near zero when the implementation is correct and would flag a real discrepancy if one existed – restoring the check's actual diagnostic power.

2.7 Grid-resolution artifact and the grid_N fix

A specific, separate numerical defect was found and fixed during this project's own development, independent of the physics above: the ASM propagation grid resolution N_{grid} defaults to 2048 in `stage4_human_vision/pipeline.py`'s `run_stage4_point` function (verified: line `grid_N=2048` in the function signature, current file). At this resolution, computed blur radii for *large* apertures ($\gtrsim$ 1500–2000 μm) come out approximately **2× too large** compared to the converged value at $N_{\text{grid}} = 4096$ – a genuine numerical under-resolution artifact, not a physical effect. `new_sweeps/run_blur_sweep.py` was corrected to pass $N_{\text{grid}} = 4096$ explicitly on 2026-08-16; the acuity-model sweep script, `new_sweeps/run_acuity_sweep.py`, called the same underlying function but was *not* updated to match until 2026-09-12/13, leaving every acuity-model number computed from that script stale for nearly a month. This specific staleness episode, its full blast-radius accounting, and its remediation are documented as a worked case study in Section 10.4.

2.8 Validation of the ASM engine

At the closed-form-tractable limit ($d_{\text{gap}} \to 0$, $M = 0$), the pipeline output was checked directly against the exact Fraunhofer diffraction result for a circular aperture:

$$I(r) = I_0 \left(\frac{2J_1(x)}{x} \right)^2, \qquad x = \frac{\pi a r}{\lambda v_r}, \qquad EE(x) = 1 - J_0^2(x) - J_1^2(x) \tag{2.9}$$

Agreement is within 2% on the 50%- and 90%-encircled-energy radii. The 80%-radius check needs 8% tolerance: the first Airy-pattern null sits at $x = 3.8317$, where $EE \approx 0.838$ – almost exactly the 80% level, on a flat part of the encircled-energy curve where a small energy error maps to a disproportionately large radius error. Repeating the check at grid sizes from 1024 through 8192 never brought this residual below 2.3–4.6%, confirming it is genuine ill-conditioning of that specific percentile at that specific aperture regime, not an under-resolved grid (source: `MASTER_HANDBOOK.tex` Chapter 6, "Validation against the closed form").

Separately, the on-axis intensity was checked against the closed-form Fresnel-zone oscillation $I(0, z) = 4I_0 \sin^2(\pi N_F/2)$ and matched at correlation 0.997 – direct confirmation that the non-monotonic Strehl-ratio behaviour this project's sweeps show (multiple local maxima as aperture is swept at fixed M) is real Fresnel-zone interference, not numerical noise, and is one of the reasons the RSS model (Section 2.3), which is smooth and monotonic by construction, cannot represent this project's own actual blur behaviour.

2.9 Blur metrics and where each one breaks

Metric	Measures	Failure mode
$r_{EE}(50\%)$	Radius enclosing half the retinal energy	Stable and monotonic in aperture, but overestimates clinical acuity loss by roughly 7× against measured clinical data (Chapter 9)
FWHM	Width of the central diffraction/blur core only	Tracks clinical acuity loss more closely ($\sim 1.6\times$) but is *discontinuous* in the design variable: a jump from 4.7 μm to 154.7 μm was observed between two apertures only 250 μm apart at $M = 4$D, from ring-hopping between local intensity maxima
Strehl ratio	Peak intensity relative to the diffraction limit	Genuinely multi-peaked (6–7 local maxima across the aperture sweep at fixed M), confirmed *worse* (more, not fewer, peaks resolved), not numerically better, at 10× finer grid sampling – i.e. this is real physics (matched to the Fresnel-zone closed form above), not numerical noise to be smoothed away

Why $r_{EE}(50\%)$ is used as the project's standard optimization metric anyway

Per individual aperture, FWHM is unusable as an optimization target because it is discontinuous in the very variable being optimized. Across a coarse sweep its *trend* tracks clinical reality more closely than $r_{EE}(50\%)$, which is smooth and numerically well-behaved but systematically pessimistic by roughly 7×. This project reports $r_{EE}(50\%)$ as its blur-optimization metric throughout and states its known bias explicitly, rather than optimizing against a metric that is discontinuous in the design variable it is meant to optimize.

2.10 Deprecated formulas

Formula	Status and measured error
$r_{\text{geo}} = \frac{a_{\text{hole}}}{2}\frac{\lvert v_r - v_f\rvert}{v_f}$	Deprecated. Superseded by the direct ASM chain (Section 2.4), which produces this effect automatically without a separate geometric-blur term.
$r_{\text{tot}} = \sqrt{r_{\text{geo}}^2 + r_{\text{diff}}^2}$ (RSS, Section 2.3)	Deprecated. 64.8% mean, 192.8% maximum divergence from the exact result.
$\frac{a_{\text{eff}}}{\sqrt{a_{\text{hole}}^2 + (2.44\lambda d_{\text{gap}}/a_{\text{hole}})^2}} \approx$	Deprecated. Smooth and monotonic by construction; cannot represent the genuine Fresnel-zone oscillation confirmed in Section 2.8.

2.11 Array optics reuse single-aperture optics unmodified

Every hole in a fabricated array has optics *identical* to a standalone single aperture of the same diameter a_{hole} at the same myopia power M: the array geometry changes only whether zero, one, or multiple holes' images simultaneously reach the retina (blinding, clean imaging, or ghosting – Chapter 5), never the per-hole blur itself. This is why every array evaluation result in this paper (Chapters 6–7) looks up its predicted acuity directly from the single-aperture sweep at the array's own chosen aperture, rather than re-deriving acuity per array family – a claim independently confirmed in this paper's own analysis (Section 7.6): all six array evaluation models tested produce numerically identical predicted LogMAR per myopia power, differing only in which *geometry* carries that identical optical aperture.

Chapter 3

The Acuity Model

A blur radius alone is not the quantity a clinician, or an IYPT judge, ultimately cares about: the relevant output is predicted *visual acuity*, conventionally reported in LogMAR (log Minimum Angle of Resolution) or Snellen notation. This project converts the ASM-computed retinal blur into a LogMAR prediction via a specific, cited pipeline implemented in `stage4_human_vision/acuity.py` (verified read directly).

3.1 Why blur radius is the wrong optimization target on its own

Minimizing a blur-radius metric directly (e.g. $r_{EE}(50\%)$) does not necessarily minimize predicted acuity loss, because the mapping from blur to acuity is nonlinear and itself depends on the eye's own optical transfer function and on retinal illuminance, not on blur radius alone. `MASTER_HANDBOOK.tex` Chapter 5 documents two successive corrections made to an earlier, blur-radius-only optimization approach: first an explicit brightness floor (Section 3.4), then optimizing acuity directly rather than blur (used throughout this project's array-evaluation chapters).

3.2 Pupil and retinal illuminance

Retinal illuminance is computed from the light entering through the active aperture(s) at a given gaze condition, scaled by scene luminance, and reported in trolands (Td), the standard unit for retinal illuminance referenced to pupil area. This project's default scene luminance is 10,000 cd/m^2 (bright outdoor daylight), with default field diameter 20° and default observer age 28.58 years (source: `stage4_human_vision/config.py` defaults, `default_scene_luminance_cdm2`, `default_field_diameter_deg`, `default_age_years`; verified present).

3.3 MTF, contrast sensitivity, and Visual Strehl

The acuity predictor uses the aperture's actual modulation transfer function (MTF, derived from the ASM-computed point spread function), evaluated against a contrast sensitivity function (CSF) model, and a Visual Strehl ratio (VSOTF) – the CSF-weighted similarity of the actual optical transfer function to the diffraction-limited one – as an overall optical-quality summary statistic. This is the same VSOTF/CSF machinery discussed in detail, with a documented correction, in the array-level ghosting theory (Section 5.4); the acuity model here uses `stage3_human_vision`'s own optical-quality and CSF evaluation directly, not a re-implementation.

3.3.1 The recommended predictor

$$\mathrm{MAR}_{10} = \frac{f_{20/20}}{f_{\mathrm{cutoff},10\%}} \qquad \text{(MAR from the MTF's 10\% cutoff spatial frequency)} \tag{3.1}$$

$$\mathrm{MAR}_{\mathrm{VSOTF}} = \frac{\mathrm{MAR}_{10}}{\mathrm{clip}(\mathrm{VSOTF}, 0.25, 2.0)^{0.1}} \qquad \text{(Visual-Strehl adjustment)} \tag{3.2}$$

$$\mathrm{LogMAR}_{\mathrm{raw}} = \log_{10}\big(\max(\mathrm{MAR}_{\mathrm{VSOTF}}, 10^{-6})\big) \tag{3.3}$$

This is the `mtf10_vsotf_brightness_adjusted` model (source: `stage4_human_vision/acuity.py`, function `predict_acuity`, verified read directly), the "recommended" model this project selects by configuration, not chosen post-hoc per result.

A legacy alternative predictor (an uncited $\mathrm{SNR}^{0.05}$ correction term, in the same file, function argument `legacy_mar`) is retained in the codebase for direct comparison, per this project's own stated design principle of keeping alternative models available rather than silently deleting a superseded one (`DESIGN.md`, cited in `acuity.py`'s own module docstring).

3.4 The brightness correction

An additive LogMAR penalty is applied in the log domain for retinal illuminance below a literature-cited knee:

$$\Delta\mathrm{LogMAR}_{\mathrm{brightness}} = \begin{cases} 0 & E_{\mathrm{td}} \geq E_{\mathrm{knee}} \\ s \cdot \log_{10}\left(\dfrac{E_{\mathrm{knee}}}{E_{\mathrm{td}}}\right) & E_{\mathrm{td}} < E_{\mathrm{knee}} \end{cases} \tag{3.4}$$

with knee $E_{\mathrm{knee}} = 300\,\mathrm{Td}$ and slope $s = 0.060$ LogMAR per $\log_{10}$-unit of illuminance (source: `stage4_human_vision/acuity.py`, `brightness_logmar_correction`, verified read directly).

Knee value (300 Td): cited to Van Nes & Bouman (1967) [Van Nes and Bouman, 1967] – threshold modulation (contrast sensitivity) is illuminance-*independent* (Weber–Fechner behaviour) above 300 Td, and increases as $E_{\mathrm{td}}^{-1/2}$ (de Vries–Rose behaviour) below it. Verified directly against the primary abstract per this project's own citation-verification discipline (`stage4_human_vision/literature.py`).

Slope (0.060 LogMAR/$\log_{10}$-unit): cited to Tidbury, Czanner & Newsham (2016), verified directly against the full text (their Table 3, $p < 0.001$), and cross-checked (not directly used) against Sheedy, Bailey & Raasch (1984)'s independently-derived value of ≈ 0.0664, which could only be verified against a secondary summary rather than the primary source – the directly-verified Tidbury value is the one used, per this project's stated preference for primary-source-verified citations over secondary ones of comparable magnitude.

Rabin (1994) is cited as independent corroboration of the qualitative direction of this effect (visual acuity is considerably *more* resistant to reduced luminance than contrast sensitivity is, in the photopic range they tested: a $116 \to 0.23\,\mathrm{cd/m^2}$ reduction gave a 3× acuity loss against a 17× contrast-sensitivity loss) – but the exact numerical knee and slope used in Equation 3.4 come from Van Nes & Bouman and Tidbury et al. respectively, both natively reported in the units this equation needs (trolands, LogMAR), not from Rabin's own $\mathrm{cd/m^2}$-based data.

Every design evaluated in this project's array chapters (Chapters 6, 7) sits well above the 300 Td knee at the default $10{,}000\,\mathrm{cd/m^2}$ scene luminance; the brightness term is therefore **inert (zero)** for every reported result in this paper, and is documented here in full for completeness and falsifiability, not because it changes any headline number.

3.5 Full acuity-model input inventory

Input	Default value	Source
Scene luminance	10,000 cd/m^2	`config.py`
Field diameter	20°	`config.py`
Observer age	28.58 years	`config.py`
Pupil diameter	4000 μm	project-wide constant
ASM grid resolution	4096 (post-fix, Section 2.7)	`run_acuity_sweep.py`
Wavelength	550 nm (vacuum)	`run_acuity_sweep.py`
Spectacle gap	12 mm	`run_acuity_sweep.py`

Chapter 4

Single-Aperture Experimental Validation

Three independent single-aperture experimental programmes were run against the theory of Chapter 2: a photographic scenery test scored by a no-reference image quality metric (BRISQUE), an optical-bench defocus sweep using a frosted projection screen, and a LogMAR acuity chart protocol. All raw data referenced below is read directly from its own CSV; no number in this chapter is retyped from a prior summary.

4.1 BRISQUE scenery test

Fifty raw DNG photographs were captured of the same outdoor scene through seven conditions: six pinhole apertures (400, 500, 700, 1000, 1200, 1400 μm) and one no-pinhole ("M1-8") control, converted to 8-bit TIFF via percentile-stretch (never JPEG, whose tone-curve compression would clip dynamic range before measurement) and scored with OpenCV's BRISQUE implementation (`cv2.quality.QualityBRISQUE_compute`), a no-reference natural-image-statistics quality metric where *lower* scores indicate a more natural-looking (less distorted) image.

Condition	Mean BRISQUE score
No-pinhole control	39.5
700 μm	54.9
1000 μm	63.8
1200 μm	67.9
500 μm	68.5
400 μm	68.3
1400 μm	69.0

(source: `ExperimentalComparison/ExP1_clarity_analysis/measurements/brisque_scores.csv`, mtime 2026-09-09; computed directly from the raw grouped means, not copied from a prior report.)

An honest, unresolved tension

Every pinhole condition scores *worse* (higher BRISQUE) than the no-pinhole control, and scores do not monotonically improve as aperture size increases toward the theoretical optimum – the opposite ordering from what the blur-radius theory (Chapter 2) predicts. BRISQUE measures general natural-image statistical "naturalness," not optical sharpness directly, and a pinhole photograph – vignetted, lower-light, geometrically different from an unrestricted exposure – departs from the natural-image statistics BRISQUE was trained against for reasons that

have nothing to do with resolved blur. This is reported here exactly as found: BRISQUE is not treated in this paper as a validated proxy for the blur-radius theory, and this discrepancy is not smoothed over.

4.2 Frosted-screen optical-bench defocus sweep (ExP3)

A fixed pinhole (one of six apertures: 400, 500, 600, 800, 1000, 1400 μm, plus an unrestricted-beam control) illuminates a frosted projection screen slid along the optical axis through five displacements (0, 5, 10, 15, 20 mm); the illuminated spot's radius is measured photographically at each displacement via blue-LED point-source imaging, using BT.601 luma (`cv2.cvtColor(..., COLOR_BGR2GRAY)`) rather than a naive max(R,G,B) channel combination – the correct convention for a blue-dominant point source, confirmed during this project's own analysis.

Condition	Growth slope (px/mm displacement)
No-pinhole control (unrestricted beam)	10.71
1000 μm	0.88
400 μm	0.72
1400 μm	0.68
500 μm	0.66
800 μm	0.65
600 μm	0.60

(source: `ExperimentalComparison/ExP3_frosted_screen/measurements/all_spot_measurements.csv`; slopes computed directly from a linear fit of measured spot radius against screen displacement per condition, this analysis.)

The unrestricted-beam control's spot grows roughly **15–18× faster** with screen displacement than any pinhole condition – direct, controlled, photographic confirmation of the stenopeic effect itself (Section 2.2): restricting the admitted ray bundle genuinely and dramatically slows how fast defocus blur grows with axial displacement, independent of any specific numeric optimum among the pinhole apertures. Among the pinhole conditions themselves the slope is not perfectly monotonic in aperture size (600 μm measured lowest, not the largest or smallest aperture tested) – a real, non-monotonic experimental result, reported as measured rather than smoothed toward the simple theoretical expectation of monotonic growth with aperture.

4.3 LogMAR acuity chart protocol

A single-pinhole logMAR acuity measurement protocol is specified in `Array_Design/array_geometry_lab/Phys` (verified present, mtime 2026-09-11): recruit subjects across a range of native myopia, measure uncorrected acuity and pinhole-corrected acuity at a fixed viewing distance, and compare the *shape* of the resulting curve (flat vs. steep against myopia severity) against the flatness prediction of Section 9.2 – a qualitative-ranking design chosen specifically because it does not require knowledge of any subject's exact refractive prescription, only a relative ranking of myopia severity, and because a flat-vs-steep *shape* comparison is a stronger test than comparing a single subject to a single predicted number.

[NEEDS VERIFICATION: As of this writing, no CSV or log file recording actual executed subject data for this protocol was found anywhere in the repository (searched: all .csv files for a "subject" column or per-subject naming, all logs, all folders named with Array-Exp/ArrayExp/clinical/trial patterns). The document

above is a *protocol specification*, not a completed experiment. Section 8.4 addresses the array-level version of this same gap explicitly.]

4.4 Reconciliation with theory

The frosted-screen result (Section 4.2) is the cleanest, most direct confirmation among the three single-aperture experiments: it isolates exactly the physical quantity the stenopeic-effect argument (Section 2.2) predicts – growth rate of blur with axial defocus – and controls for everything else (same light source, same imaging chain, same measurement method) across every condition. The BRISQUE result (Section 4.1) is reported honestly as *not* confirming the blur-radius theory in its raw form, precisely because BRISQUE is not measuring the same physical quantity the theory predicts; this is treated in this paper as a limitation of the chosen validation metric, not as evidence against the underlying optics.

Chapter 5

Theoretical Framework II: Array-Level Optics

5.1 Mathematical array representation

An array design is fully specified by: a *family* (the geometric rule generating hole center positions $\{(x_i, y_i)\}$), a hole diameter a_{hole} (identical for every hole in a given design), a requested hole count N, and family-specific spacing parameters (e.g. pitch and rotation for a lattice family, ring spacing and holes-per-ring for a ring family). Eleven families are implemented in this project's generator library (`Array_Design/array_geometry_lab/array_calculation/generators/`, verified via directory listing): `square`, `hexagonal`, `rectangular`, `concentric`, `concentric_centered`, `ring_offset_concentric`, `archimedean_spiral`, `fibonacci_spiral`, `golden_angle`, `poisson_disk`, and `jittered_lattice`. Every generator recenters its own output on its own centroid before returning coordinates (`base_generator.py`, shared step), so every design is symmetric about the array's own optical axis by construction.

For a square lattice with pitch p and rotation θ_0:

$$(x_i, y_i) = R(\theta_0) \cdot p\,(m, n), \qquad (m, n) \in \mathbb{Z}^2 \tag{5.1}$$

For a hexagonal lattice with the same pitch p (nearest-neighbor spacing):

$$(x_i, y_i) = R(\theta_0) \cdot p\left(m + \tfrac{1}{2}n,\ \tfrac{\sqrt{3}}{2}n\right), \qquad (m, n) \in \mathbb{Z}^2 \tag{5.2}$$

Both are checked directly against their own regenerated coordinates in Section 7.5 against real fabricated designs.

5.2 Fill factor and brightness

The array's fill factor (fraction of the frame area occupied by open aperture) is $f = N\pi(a_{\text{hole}}/2)^2/A_{\text{hull}}$, where A_{hull} is the convex-hull area of the hole pattern. Because every real winning design in this project (Chapters 6, 7) operates with only one hole illuminated at a time under normal (straight-ahead) gaze (Section 2.11), the fill factor is not itself a brightness constraint in the way it would be for a simple neutral filter; it is a *geometric* density constraint, trading off against the blind- fraction risk of Section 5.6 – a sparser array is more likely to leave a gaze direction with no active hole at all.

5.3 Comprehensive ghosting theory

5.3.1 The crude formula and why it was replaced

An early criterion for this project treated adjacent holes at pitch s as producing retinal images separated by $\Delta r \approx s\, v_r / d_{\text{gap}}$, and called a design "ghost-free" whenever Δr was smaller than the single-hole blur radius. Checked against two real, sold multi-pinhole eyewear patents (KR 10-2081511, pitch 3.2–4.8 mm; CN 201716487U, pitch 0.7–1.2 mm; both $d_{\text{gap}} \approx 12$ mm), this formula predicted severe (30×–400×) violations for *both* sold, presumably-functional products – the contradiction that motivated a full re-derivation (source: `Array_Design/ghosting/THEORY.md` Section 1, verified read directly). The formula was subsequently found to be independently wrong on its own terms (re-derivation, `THEORY.md`'s own addendum) and is used nowhere in this paper's reported results; it is included here only as the historical motivation for the derivation that follows.

5.3.2 Multi-hole image formation

Under ordinary incoherent daylight illumination, the mutual coherence between light paths through macroscopically-separated holes is negligible (Goodman, *Statistical Optics*); *intensities*, not fields, add. Each active hole produces its own copy of the single-hole point spread function, shifted to its own retinal position:

$$I_{\text{retina}}(\mathbf{r}) = \sum_i I_{\text{single}}(\mathbf{r} - \mathbf{r}_i) = I_{\text{single}}(\mathbf{r}) * \text{Comb}_{\{\mathbf{r}_i\}}(\mathbf{r}) \tag{5.3}$$

i.e. the multi-hole retinal image is exactly the single-hole PSF convolved with a Dirac comb (or, for a general non-lattice array, a general sum of shifted delta functions) at the active holes' own retinal image positions.

5.3.3 The Fourier consequence the crude formula omitted

By the convolution theorem, Equation 5.3 in frequency space is a *product*:

$$\hat{I}_{\text{retina}}(f) = \text{MTF}(f) \cdot \hat{\text{Comb}}_{1/\Delta r}(f) \tag{5.4}$$

The periodic ripple's fundamental frequency $f_0 = 1/\Delta r$ is attenuated by the single-hole MTF (already computed by this project's own ASM pipeline, Chapter 2) at that same frequency. A dense array (small pitch s, large f_0) pushes the ghosting ripple's fundamental past the frequency where the MTF has already rolled off toward zero – multiple shifted copies smooth into one continuous blur regardless of how large the raw pitch-vs-blur-radius ratio of the crude formula would suggest. This is the central reason the crude formula was wrong for *dense* designs: it never accounted for the single-hole MTF's own low-pass character at all.

5.3.4 The discrete precondition

If the number of simultaneously-active holes $N_{\text{eff}} \leq 1$ for a given pupil alignment, there is only one term in the sum of Equation 5.3 – no periodic structure exists to be attenuated or perceived, regardless of pitch. This is the *active-pairs* mechanism this project's `margins/active_pairs.py` and the ghost gates of Chapter 6 implement directly, and explains why a sufficiently sparse array (e.g. the KR patent's own pitch) rarely ghosts even though its raw pitch would fail the crude formula badly: most of the time, no second hole is simultaneously active to ghost against at all.

5.3.5 A refined three-regime criterion

Combining the two mechanisms above, this project's synthesized ghosting-perceptibility criterion is:

$$\text{Imperceptible if} \quad N_{\text{eff}}^{(\text{worst-case})} \leq 1 \quad \text{(Regime A)} \qquad \text{OR} \qquad \text{MTF}(f_0) < \text{CSF}^{-1}(f_0) \quad \text{(Regime B)} \tag{5.5}$$

Regime C – intermediate pitch, N_{eff} small but > 1, and f_0 not yet high enough for the MTF roll-off to suppress it below the CSF threshold – is the predicted *danger zone*: a candidate physical mechanism for the double/triple-vision and honeycomb/grating symptoms Kim et al. (2017) [Kim et al., 2017] directly report in real subjects wearing real multi-pinhole glasses (*"discomfort from after-images such as honeycomb images, horizontal grating, and double or triple vision"*, quoted directly, significantly worse than for single-pinhole glasses in the same study) – overturning an earlier framing of this project's own investigation, which had asked why real sold products show no reported ghosting at all; they do.

5.4 The heuristic-CSF-out-of-range bug and the Barten (1999) fix

Evaluating Equation 5.5 requires a CSF model. This project's own first-pass CSF implementation (`Array_Design/ghosting/contrast.py`) is a heuristic, peak-matched to reported CSF peaks at 4–8 cycles/degree (cpd), and was **never fit or validated below its own peak frequency**.

Pass 1: a physically impossible result, traced to its root cause

A 36-point sweep (apertures $\{300, 400, 700, 1000\}\,\mu$m × pitches $\{700, 900, 1200, 1500, 2000, 2500, 3200, 4000, 4$ $M = 4$D, $d_{\text{gap}} = 12\,$mm, $\lambda = 550\,$nm, $D_{\text{pupil}} = 4000\,\mu$m, $N_{\text{grid}} = 1024$, square packing) found **zero** Regime C points – every point classified as A (16/36) or B (20/36), i.e. every dense, potentially-dangerous pitch was classified "safe." Traced to root cause: every Regime-B point had a genuine ghosting fundamental frequency $f_0 < 0.3\,$cpd (physically correct – a near-eye, millimetre-scale pitch genuinely produces a coarse, sub-1-cpd periodic pattern), evaluated against the heuristic CSF model *far outside its own fitted range*, which in that regime returned a physically impossible required-contrast threshold of 16,337 to 1,000,000 (i.e. more than 100% contrast) – trivially satisfying Regime B ("safe") regardless of the design's real MTF value. Source: `Array_Design/ghosting/RESULTS.md`, "Pass 1" section, verified read directly.

Fix: a working implementation of the real Barten (1999) contrast sensitivity model [Barten, 1999, 2003] (simplified form via the published `colour-science` Python package) was located and verified, with a stated valid range of 0.2–50 cpd – overlapping this project's actual ghosting frequencies (0.10–0.30 cpd), unlike the heuristic. Sanity-checked first (threshold 0.6%–3.6% across a 2–60° field-size range at that frequency, near-luminance-independent – physically sane behaviour, unlike the heuristic's out-of-range artifact).

Regime	Pass 1 (heuristic CSF)	Pass 2 (real Barten CSF)
A (sparse, $N_{\text{eff}} \leq 1$)	16	16 (unchanged)
B (dense, CSF says safe)	20	**0**
C (danger zone)	0	**20**

Every previously-"safe" dense point was reclassified as danger zone – including the entire tested real dense-patent pitch range (700–1200 μm). This reclassification is directionally consistent with Kim et al. (2017)'s directly-quoted clinical finding above: the theory no longer needs an unresolved gap to explain that finding away (source: `Array_Design/ghosting/RESULTS.md`, "Pass 2").

5.4.1 Pass 3: the harmonic series, not just the fundamental

Passes 1–2 tested only the fundamental ghosting frequency $f_0 = 1/\Delta r$. A finite train of N shifted PSF copies (Equation 5.3) is a genuinely periodic signal with harmonic content at every nf_0 ($n = 1, 2, 3, \dots$), not a single sinusoid. Computing the full harmonic margin profile $\mathrm{MTF}(nf_0)/\mathrm{CSF}^{-1}(nf_0)$ for $n = 1$ to 40 at every dense-pitch test point found the worst-case harmonic is **never** $n = 1$: for the representative case $a_{\mathrm{hole}} = 400\,\mu$m, pitch $= 900\,\mu$m, $M = 4$D, the fundamental-alone margin is 75.6 (already > 1, i.e. already "danger zone" under Pass 2), but the actual worst harmonic is $n = 9$, at 2.09 cpd, with margin 329.2 – **4.4× more dangerous than the fundamental alone predicted**.

Why this generalizes, not just this one case

MTF decreases smoothly and monotonically with frequency; the Barten CSF's own sensitivity peak sits in a fixed *absolute* frequency band (∼2–5 cpd) regardless of pitch. The product of a monotonically-falling curve and a fixed-position peak has its own peak at that same fixed absolute frequency – changing the array pitch only determines *which harmonic index* n happens to land near that peak, not whether one does. Confirmed by re-running the full 36-point sweep with harmonic analysis: the worst-case margin at a given aperture is essentially constant across every dense pitch tested ($\sim$ 329 for $a_{\mathrm{hole}} = 300/400\,\mu$m, $\sim$ 273 for $700\,\mu$m, $\sim$ 219 for $1000\,\mu$m), only the specific harmonic index achieving it changes. Source: `Array_Design/ghosting/RESULTS.md`, "Pass 3."

5.5 Vignetting: gaze-dependent aperture activation

As the wearer's eye rotates by eccentricity angle φ from the array's own optical axis, the pupil's center shifts laterally relative to the (head-fixed) array by

$$\Delta(\varphi) = R_{\mathrm{rot}} \sin\varphi \tag{5.6}$$

with $R_{\mathrm{rot}} = 13.5$ mm (physiological range 13–14 mm), in an azimuth θ that can, physically, be any direction the wearer happens to look (source: `Array_Design/array_geometry_lab/array_calculation/en` function `pupil_shift_um`, verified read directly). At each shifted pupil position, the illuminated area contributed by hole i (radius R_h) is the exact circle–circle overlap area between the pupil disc (radius R_p) and that hole's disc, handled in three explicit regimes with no implicit fallthrough:

$$A_{\mathrm{overlap}}(\Delta_i, R_p, R_h) = \begin{cases} \pi \min(R_p, R_h)^2 & \Delta_i \le |R_p \\ 0 & \Delta_i \ge R_p + \\ R_p^2 \cos^{-1}\!\left(\dfrac{\Delta_i^2 + R_p^2 - R_h^2}{2\Delta_i R_p}\right) + R_h^2 \cos^{-1}\!\left(\dfrac{\Delta_i^2 + R_h^2 - R_p^2}{2\Delta_i R_h}\right) & \\ \quad - \dfrac{1}{2}\sqrt{(-\Delta_i + R_p + R_h)(\Delta_i + R_p - R_h)(\Delta_i + R_h - R_p)(\Delta_i + R_p + R_h)} & \text{otherwise} \end{cases} \tag{5.7}$$

(Handbook 2, Sec. 13.2, implemented verbatim in `vignetting_overlap.py`'s `overlap_area_um2`, verified read directly.)

A floating-point boundary defect, found and fixed

Near the exact boundary $\Delta_i = R_p + R_h$, floating-point round-off on the two $\cos^{-1}$ terms in the partial-overlap branch can produce a small *negative* computed area despite the true area being exactly zero there by construction – observed concretely as $-0.0589\,\mu\text{m}^2$ at $\Delta_i = 3000\,\mu\text{m}$ for a $2000\,\mu\text{m}$ hole. The implementation clamps the square-root term's radicand to $[0, \infty)$ as a first safeguard, and clamps the final combined area to $[0, \infty)$ as an independent second safeguard closing the same gap – both present in the verified current code, not merely proposed.

Independent validation: `vignetting_overlap.py`'s copy of this formula (a deliberate copy, not an import, so the array-geometry pipeline has zero runtime dependency on the originating module's internals) was checked against its own 72-point validation dataset and reproduces it to **$4.657\times10^{-10}\,\mu\text{m}^2$ maximum absolute difference** (tolerance $10^{-6}\,\mu\text{m}^2$), with zero negative-area outputs produced – confirming the boundary-clamp fix was copied along with the formula, not left behind (source: `VALIDATION.md`, verified present, this project's own Part 1 validation gate).

5.5.1 Single-axis vs. full-azimuth sampling

This project's original vignetting evaluation (`margins/vignetting_robustness.py`) models gaze along a *single axis* only – the pupil shifts to $(\Delta(\varphi), 0)$, never at any other azimuth θ – matching the convention of the module it was derived from. This is a real, acknowledged modelling limitation (Section 6.2), not an oversight left undocumented: the module's own docstring states plainly that this convention was inherited, and Model B (Chapter 7) was built specifically to remove it.

5.6 Blinding, and the fundamental array trade-off

A gaze direction is *blind* when the total illuminated area across every hole is exactly zero: $\sum_i A_{\text{overlap}}(\Delta_i(\varphi,\theta), R_p, R_h) = 0$ – no light reaches the retina at that gaze direction at all, as distinct from a *ghosting* gaze direction, where two or more holes are simultaneously active (quantified via the active-hole-fraction threshold of Section 5.3.5). These two failure modes trade directly against each other through the array's own pitch: a denser array (smaller pitch, for a fixed hole size) is less likely to leave any gaze direction blind, because some hole is more likely to be near the shifted pupil at any given (φ, θ) – but a denser array is, by the same token, more likely to have *two* holes simultaneously within the pupil at once, i.e. more likely to ghost. This project's own real, already-computed data confirms this trade-off is not merely theoretical: at matched pitch ($4000\,\mu\text{m}$, $M = 2\text{D}$, $a_{\text{hole}} = 750\,\mu\text{m}$), a ring-based design with 8 holes per ring achieves lower blind fraction (0.03%) than its 6-holes-per-ring counterpart (0.23%) precisely by pulling ring holes closer together angularly – but pays for it with a real 7.58% time-weighted ghost fraction, against 0.00% for the 6-hole variant (Section 7.5, Figure 7.2). Hexagonal packing is shown in Chapter 7 to be the one family tested that achieves near-zero on *both* metrics simultaneously at the same matched spacing, rather than trading one against the other.

5.7 Ghosting theory: full literature review

Section 5.3 gave the ghosting derivation in compressed form. This section expands the literature review behind it in full, reproducing the citation-fidelity discipline of the source document (`Array_Design/ghosting/THEORY.md`, verified read directly, mtime 2026-08-14 00:36) exactly: every claim below is labelled **[EXACT]** (a first-principles or exact identity), **[PUBLISHED]** (a specific cited result applied as originally derived), **[SYNTHESIS]** (a novel combination of citations not itself published as a unified result), or **[UNTESTED]** (a claim requiring physical or psychophysical confirmation this project has not obtained). This labelling convention is preserved here deliberately, not smoothed into uniform prose, because it is itself part of this project's own honesty discipline: a reader should be able to tell, claim by claim, which statements in this section are load-bearing established physics and which are this project's own proposed synthesis.

5.7.1 Motivation: two real, sold products the original criterion could not explain

An early criterion for this project treated adjacent holes at pitch s as producing retinal images separated by $\Delta r \approx s\, v_r / d_{\text{gap}}$ **[EXACT, as originally labelled – see below]**, and called a design "ghost-free" when Δr was smaller than the single-hole blur radius $r_{EE}(P)$. Checked against two real, commercially sold pinhole-eyewear patents – KR 10-2081511 (pitch 3.2–4.8 mm) and CN 201716487U (pitch 0.7–1.2 mm), both with $d_{\text{gap}} \approx 12$ mm and $v_r \approx 22$–24 mm – this criterion predicted severe (30×–400×) violations for *both* presumably-functional, sold products. This direct contradiction between a theoretical prediction and two real commercial devices is what motivated the full re-derivation this section and Section 5.3 document.

A label that did not survive re-derivation, kept visible rather than quietly corrected

Equation (1) above (the $\Delta r \approx s\, v_r / d_{\text{gap}}$ form) was originally marked **[EXACT]** in this project's own working notes. An independent re-derivation subsequently found this label did not hold – the formula is used nowhere in this paper's reported results (Section 5.3 uses the corrected convolution-theorem derivation instead) – and the source document's own convention is to leave the original claim, and its original (incorrect) fidelity label, visible in place with a note pointing to the correction, rather than silently edit history. This paper follows that same convention here.

5.7.2 2.1 Lenslet-array merging theory (Gabor superlens)

The closest existing precedent for a quantitative multi-channel merging condition in the optics literature is Gabor-superlens theory (a microlens-array imaging architecture, originating 1941, with modern quantitative treatments from 2014 **[NEEDS VERIFICATION: exact citation for the specific 2014 treatment referenced in the source document was not itself recorded there beyond the year – not independently tracked down for this paper]**): mismatched, non-integer-ratio pitches in a two-array relay cause imaging channels to, in the cited work's own language, "unite to form superfocal lines much smaller in number than the number of microlenses" **[PUBLISHED]**. This is cited here as evidence that the general class of problem – multiple discrete optical channels merging or failing to merge into a single perceived image – is a solved, quantitatively tractable one in an adjacent field, *not* as a directly transferable formula: a Gabor superlens is a two-array relay forming a real intermediate image, physically a

different system from a single perforated occluder imaged directly by the eye's own optics, and no attempt is made in this project to import the Gabor merging condition as-is.

5.7.3 2.2 Direct evidence that multi-aperture ghosting is a real, reported clinical symptom

The single most important literature finding underlying this section: Kim, Park, Park & Chun (2017, *J Korean Med Sci* 32(5):850–857) [Kim et al., 2017] – already present in this project's own vetted citation library (`stage4_human_vision/literature.py`, and independently used elsewhere in this paper for a wholly separate purpose, Section 9.3) – directly reports, in real human subjects wearing real multi-pinhole glasses, "discomfort from after-images such as honeycomb images, horizontal grating, and double or triple vision" **[PUBLISHED, direct quotation from the primary source]**, significantly worse than for single-pinhole glasses tested in the same study. This finding overturns the original framing of this project's own investigation: the question this project initially set out to answer was "why do real, sold multi-pinhole products show no reported ghosting problem" – and the honest literature answer is that they *do* report exactly this problem, in a directly-quoted clinical source. The question this section's derivation (Section 5.3) actually needed to answer is therefore not "why is ghosting absent," but "why do the two specific sold patents checked in Section 5.3 not appear to address it explicitly in their own documentation, and under exactly what geometric conditions does the symptom become severe."

5.7.4 2.3 An independent real-world statement of the discrete-activation mechanism

A separate, unrelated US patent (cited in the source document by its own title, "Method for facilitating pinhole effect..." **[NEEDS VERIFICATION: full patent number not recorded in the source THEORY.md document beyond this title fragment – not independently tracked down for this paper]**) states, entirely independently of anything already in this project's own derivation: "If two pinholes are separated by less than the diameter of the pupil aperture, two pencils of ray… form two retinal images. Further, too large of a separation will result in dead spots… while too small… will produce multiple images." **[PUBLISHED, direct quotation]**. This is significant specifically because it states a *discrete precondition* – whether two or more holes are simultaneously illuminating the pupil at all – rather than a continuous margin comparison. This is exactly the N_{eff} quantity this project's own active-pairs machinery (`margins/active_pairs.py`, Section 5.3.5) already computes via direct discrete-lattice enumeration, independently of this patent. The connection between the patent's own stated mechanism and this project's own pre-existing N_{eff} computation was not recognized until this literature review was written – i.e. this project's own Regime-A criterion ($N_{\text{eff}} \leq 1$, Equation 5.5) turns out to be independently corroborated by real patent-literature language describing the identical physical mechanism, discovered after the fact rather than designed to match it.

5.7.5 2.4 Periodic-artifact visibility: the screen-door-effect precedent

Display engineering has a standard, mature treatment of an analogous problem: the visibility of a display's own pixel/subpixel grid structure (the "screen-door effect"), which is assessed by expressing the grid's periodicity in cycles per degree of visual angle, determining the resulting pattern's modulation depth, and comparing that modulation depth against the human contrast sensitivity function's own threshold at that spatial frequency **[PUBLISHED]**. This confirms that the three-regime criterion's own frequency-domain framework (Section 5.3, Equation 5.5) is the same established methodology display engineering already uses for a structurally analogous

periodic-artifact-visibility problem, not a bespoke construction invented for this project alone. The same literature also confirms that a cruder, acuity-only rule (of the same general type as this project's own original, later-superseded criterion, Section 5.3) is a common, recognized first-pass approximation in that adjacent field too – i.e. this project's own original mistake was not a mistake unique to this project, but a documented, recurring simplification in the broader periodic-visual-artifact literature, corrected here by the same frequency-domain methodology that field already uses to correct it.

5.7.6 Synthesis: the three-regime criterion, and what remains untested

Combining 2.2's discrete-activation precondition, 2.3's independent confirmation of the same mechanism, and 2.4's frequency-domain visibility methodology, with this project's own convolution-theorem derivation (Section 5.3), yields Equation 5.5's three-regime criterion **[SYNTHESIS]** – a novel combination of these separately-published pieces, not itself a result published as a unified criterion anywhere in the cited literature. The predicted danger-zone mechanism (Regime C) is directionally consistent with Kim et al.'s directly-quoted clinical symptom report (2.2 above), but this consistency is *directional*, not a quantitative validation: this project has not obtained psychophysical confirmation that the specific Barten-CSF threshold crossing computed in Section 5.4–5.4.1 predicts *which specific* real designs a real wearer would report symptoms for, only that the general mechanism and its general conclusion (dense real patent pitches are more dangerous than a heuristic CSF model suggested) point the same direction as the one directly-quoted clinical source available **[UNTESTED]** – stated here explicitly as a limitation of this specific sub-claim, in addition to the broader limitations already catalogued in Chapter 12.

5.8 Full generator equations for all eleven families

Section 5.1 gave the two most-used families (square, hexagonal) in full. This section derives all eleven implemented families (`array_calculation/generators/`, verified read directly file-by-file; mtimes in Chapter 14), since Chapters 6–7 evaluate every one of them and Section 7.5's symmetry argument only has force once the alternatives are actually written down.

Every generator shares one post-processing step, `base_generator.finalize()` (verified read directly): greedy overlap rejection (points are kept in generation order, any later point closer than a_{hole} to an already-kept point is discarded), then recentering on the surviving set's own centroid. This shared step is the reason a truncated (non-lattice-complete) candidate set can end up off-centre – the mechanism behind Section 5.8.6 below – and is stated once here rather than repeated in every family's own derivation.

5.8.1 Square

$$(x_{m,n}) = R(\theta_0) \cdot p\,(m, n), \qquad (m, n) \in \mathbb{Z}^2 \tag{5.8}$$

Free parameter: pitch p (nearest-neighbor spacing), plus optional rotation θ_0. Construction: a square meshgrid large enough to contain N points, radially sorted by distance from the origin, the closest N kept, then rotated by θ_0 (rotation is applied *after* the radial sort, so it changes only each point's (x, y), never which points are selected – rotation preserves every pairwise and every origin distance exactly).

5.8.2 Hexagonal

$$y_r = r\,p\,\frac{\sqrt{3}}{2}, \qquad r \in \mathbb{Z} \tag{5.9}$$

$$x_{r,m} = mp + \begin{cases} p/2 & r \text{ odd} \\ 0 & r \text{ even} \end{cases}, \qquad m \in \mathbb{Z} \tag{5.10}$$

Row height $p\sqrt{3}/2$ and alternate-row half-pitch offset are the standard hexagonal close-packing construction; identical radial-sort-then-keep-closest-N selection as square.

5.8.3 Rectangular

$$(x_{m,n}) = R(\theta_0) \cdot (mp_x, np_y), \qquad (m,n) \in \mathbb{Z}^2 \tag{5.11}$$

Independent x- and y-pitches p_x, p_y. At $p_x = p_y$ this reduces to *exactly* the same lattice as square's own construction – the two generators use deliberately structurally identical code (same meshgrid-then-radial-sort approach) specifically so this equivalence holds exactly, not approximately, and is checked directly in `spot_check_generators.py`.

5.8.4 Concentric

Ring $k = 0$ is a single hole at the origin. Ring $k \geq 1$ sits at radius $r_k = k\,s$ (ring spacing s) with

$$n_k = \text{round}(n_0\,k), \qquad \theta_{k,j} = \frac{2\pi j}{n_k},\ j = 0, \ldots, n_k - 1 \tag{5.12}$$

hole count *scaling with ring circumference* (constant angular spacing between adjacent holes on every ring, equal to ring 1's own spacing $2\pi/n_0$) rather than a fixed count per ring – a fixed count would make outer rings implausibly sparse or force inner-ring holes closer than a_{hole} (which `finalize()`'s overlap rejection would then discard anyway), so the constant-angular-spacing rule is the more physically sensible default for a fabricatable pattern (source: module docstring, this exact reasoning, verified read directly).

5.8.5 Ring-offset concentric

Identical ring radius/count formula to Section 5.8.4, with one addition: odd-indexed rings (k odd) receive a fixed half-angular-step rotation π/n_k relative to their unrotated position; even rings receive none –

$$\theta_{k,j} = \frac{2\pi j}{n_k} + \begin{cases} \pi/n_k & k \text{ odd} \\ 0 & k \text{ even} \end{cases} \tag{5.13}$$

the "brick-course" pattern: adjacent rings' holes sit staggered rather than radially aligned. An *alternating* offset (not a progressively-accumulating one) was chosen as the more literal reading of a brick pattern – real brick courses alternate, they do not progressively drift – and because it is the simpler, unambiguous construction; a progressive variant was considered and explicitly not implemented, per the module's own stated reasoning against shipping an unused second flag.

5.8.6 Concentric-centered, and a real, documented bug it exists to fix

Investigating this family's own motivation directly (rather than assuming it), its author found that `concentric.py` and `ring_offset_concentric.py` *already* start their own point list with a literal centre hole at $(0, 0)$ – so the reason neither family's *final*, post-`finalize()` output has a hole at the origin is not a missing centre point, but `base_generator`'s own shared recentering

step: a *complete* ring sums to exactly $(0, 0)$ by rotational symmetry and never disturbs this, but a *truncated* final ring (the common case, whenever the requested N does not exactly fill an integer number of complete rings) does not sum to zero, and the shared recenter-on-centroid step then shifts the nominal centre hole away from the true origin by exactly that leftover imbalance.

Confirmed against a real, already-fabrication-relevant case

`ring_offset_concentric`'s own $N = 20$, ring spacing $4000\,\mu$m, $n_0 = 8$ candidate – its real $M = 2$–3D winner in an earlier evaluation pass – has its nominal centre hole recentred to $1704.8\,\mu\mathbf{m}$ from the true origin. Reproduced bit-for-bit by a naive "prepend then recenter together" re-implementation, confirming this mechanism directly, not by inference.

Fix, and the one substantive difference this generator adds: recenter the ring points on their *own* centroid first, *then* prepend the $(0, 0)$ centre hole, *then* run `finalize()` once more on the combined set. Since the ring points are already exactly zero-mean before the centre hole is added, and the centre hole itself contributes $(0, 0)$ to any mean, the combined set's mean is exactly $(0, 0)$ by construction – the final recentering step becomes provably a no-op. Checked directly on the identical displaced case above: centre-hole displacement drops to $1.6 \times 10^{-13}\,\mu$m (floating-point noise), confirming the fix, not merely asserting it.

5.8.7 Archimedean spiral

$r = c\theta$ (unwrapped, not reduced mod 2π) – the curve where each full revolution increases r by the constant amount $2\pi c$. Points are placed along the curve at approximately constant arc-length spacing: since $ds/d\theta = \sqrt{r^2 + c^2}$ for this curve, a first-order (Euler) angular step

$$d\theta = \frac{a_{\text{hole}} \cdot f_{\text{spacing}}}{\sqrt{r^2 + c^2}} \tag{5.14}$$

(evaluated at the current r before each step, $f_{\text{spacing}} = 1.5$ by default) keeps successive points near-evenly spaced along the arc. Explicitly *not* the same construction as the Fibonacci/golden-angle family below – this generator advances along one continuous outward-spiralling arm; the golden-angle family jumps by a fixed large irrational angle each step and never traces a continuous curve at all. Confirmed different point clouds directly, not merely by differing formulas (`spot_check_generators.py`'s archimedean-vs-fibonacci difference check).

5.8.8 Fibonacci (Vogel) spiral and golden-angle phyllotaxis

$$r_n = c\sqrt{n}, \qquad \theta_n = n\,\alpha_g, \qquad \alpha_g = \pi(3 - \sqrt{5}) \approx 137.50776^\circ, \qquad n = 1, \ldots, N \tag{5.15}$$

$\alpha_g = 2\pi(1 - 1/\varphi)$, φ the golden ratio – the irrational angle that maximizes packing uniformity for this family (Vogel, 1979). "Golden-angle phyllotaxis" and the "Fibonacci/Vogel spiral" are, mathematically, the *identical* formula – there is no separate standard point-packing construction in the literature distinct from this one (the visually different "nested-squares Fibonacci spiral" is an unrelated aesthetic curve, not a packing formula, and is not what either generator name refers to here). Rather than risk two copies silently drifting apart under future edits, `golden_angle.py` re-exports `fibonacci_spiral.generate()` directly under its own family label; for the same (N, a_{hole}, c) the two generators' output coordinates are, by construction, identical.

5.8.9 Poisson-disk (blue-noise)

Stochastic; minimum pairwise distance $r = a_{\text{hole}} f_{\text{spacing}}$ enforced via Bridson's algorithm (Bridson, SIGGRAPH 2007 sketches) rather than naive dart-throwing-with-rejection, whose rejection rate is unbounded as the domain fills (each new random dart becomes increasingly likely to land too close to an existing point). Bridson's method instead grows the set from an active frontier – only ever proposing new candidates in the annulus $[r, 2r]$ around an existing active point, checked against neighbours via a background grid of cell size $r/\sqrt{2}$ (small enough that each cell holds at most one sample) – and terminates deterministically once no active point can produce a valid candidate after $k = 30$ tries, giving $O(1)$ amortized work per accepted or retired point rather than the unbounded degradation of naive rejection sampling. The domain is filled to near-maximal packing, then the N points closest to the origin are kept (same selection convention as square/hexagonal). Because the algorithm is stochastic, `seed` is a required, not optional, parameter for every call to be reproducible.

5.8.10 Jittered lattice

Starts from `hexagonal.generate()`'s own lattice, then perturbs every hole by a random offset sampled *uniformly by area* within a disk of radius $f_{\text{jitter}}\, p$:

$$\rho = f_{\text{jitter}}\, p \sqrt{u}, \qquad u \sim \text{Uniform}(0, 1), \qquad \phi \sim \text{Uniform}(0, 2\pi) \tag{5.16}$$

The $\sqrt{u}$ is load-bearing, not optional: sampling the offset magnitude directly as $f_{\text{jitter}}\, p\, u$ would over-concentrate perturbed points near zero offset, since area scales as r^2, not r – the same disk-sampling construction already used elsewhere in this project's own Monte Carlo validation work (`array_optimization/vignetting.py`). A *uniform*, not Gaussian, offset distribution was chosen specifically because the parameter's own contract is "offset up to $f_{\text{jitter}}\, p$" – a hard, stated bound; a Gaussian has unbounded support, which would make that "up to" contract false in the tail, however unlikely. The jittered set is then re-run through `finalize()`'s own overlap rejection (jitter can push two holes that were exactly at the lattice's own valid spacing closer than a_{hole} – these must be rejected exactly as any other overlapping candidate would be, not silently kept because they originated from a valid lattice). At $f_{\text{jitter}} = 0$ every offset is identically zero, so this reduces exactly to `hexagonal.generate()`'s own output – checked directly, not merely argued, in `spot_check_generators.py`.

Chapter 6

Evaluation Model A: The Single-Axis Worst-Case Gate

Model A is this project's original array-safety evaluation: for each candidate design, apply three independent pass/fail gates, then rank surviving candidates by packing density. It was this project's primary standard until an explicit methodology change on 2026-09-13 (Chapter 7) promoted Model B in its place; Model A's own results remain valid and are still fully reported here, not deleted, per this project's own stated policy of keeping superseded models available for comparison.

6.1 The three gates

Implemented in `array_evaluation/comparison_metric/gates.py` (verified read directly, current file):

$$\text{Ghost safety:} \quad \text{worst_case_ghost_margin} \geq \frac{2.0 \times 2.9297\,\mu\text{m}}{\text{resolvable_blur_um}} \tag{6.1}$$

$$\text{Vignetting safety:} \quad \text{worst_case_vignetting_margin} \geq \frac{2.0 \times 2.9297\,\mu\text{m}}{\text{resolvable_blur_um}} \tag{6.2}$$

$$\text{Viability:} \quad \text{min_illuminated_area_across_}\varphi \geq 0.20 \times \pi\left(\frac{a_{\text{hole}}}{2}\right)^2 \tag{6.3}$$

The margin threshold's numerator, $2.9297\,\mu$m, is $2\times$ the ghost-formula validation grid's own pixel pitch ($1.46\,\mu$m – the same measurement floor noted in Section 5.5), i.e. twice the finest resolution at which the underlying ghost-separation formula was itself validated; the $2.0\times$ safety multiple on top of that is applied per this project's own stated design brief, not independently derived. The viability floor (20% of one hole's own full area) requires that floor to hold at *every* tested gaze angle φ, not merely at $\varphi = 0$ – an audit fix documented in `gates.py`'s own module docstring as replacing an earlier, weaker "margin > 0" check with this real, sourced threshold.

6.2 The single-axis limitation, acknowledged

Model A's ghost and vignetting margins are computed via `margins/vignetting_robustness.py`, which samples gaze along a *single axis* only ($\theta = 0$ always – Section 5.5.1). This is a genuine limitation, not merely a simplification: for a ring-based family in particular, a complete blackout can occur in the angular *gap* between two ring holes at one azimuth while an adjacent azimuth at the identical eccentricity sits directly on top of a hole – single-axis sampling can silently miss this. A concrete, real instance of exactly this failure mode was found in this project's own development:

a `concentric_centered` candidate that outperformed the then-current winner everywhere inside 15° failed Model A's worst-case safety gate specifically via a complete blackout in the 20–25° gaze range (source: `Array_Design/array_geometry_lab/array_data/evaluation_results/CENTERED_CONCEN` 2026-08-17, verified present) – the concrete, real-world motivating example for building Model B's full-azimuth sampling (Chapter 7).

6.3 Current results

Winner per myopia power, ranked by packing density among gate survivors (source: `Array_Design/array_geome` mtime 2026-09-13 00:42 – see Chapter 14 for the full provenance record):

M (D)	Family	a_{hole} (μm)	N	Worst-case margins
0	square	2000	9	∞/∞
1	square	1250	100	∞/∞
2	square	750	100	∞/∞
3	square	750	100	∞/∞
4	square	600	50	∞/∞
5	square	600	50	∞/∞
6	square	500	50	∞/∞
8	square	400	50	∞/∞

An infinite margin means zero simultaneously-illuminated hole pairs at every single-axis-tested gaze angle – a structural, geometric guarantee under the single-axis sampling convention, not a narrowly-cleared numeric threshold. This structural safety is exactly why Model A alone did not surface the real ring-family blackout of Section 6.2: a design can be structurally safe along the one tested axis while still failing badly off it.

Chapter 7

Evaluation Model B: Two-Dimensional Gaze-Weighted Evaluation

7.1 Methodology change

On 2026-09-13, following an explicit project decision, gaze-weighted (time-weighted ghost/blind fraction) evaluation was promoted from a secondary, comparison-only analysis to this project's **primary** array-safety standard, judged more realistic than a pure worst-case-across-every-tested-gaze-angle criterion (source: `Array_Design/array_geometry_lab/GAZE_WEIGHTED_APPROACH.md`, "UPDATE, 2026-09-13" section, and `array_data/evaluation_results/TIMELINE.txt`, both verified present, mtime 2026-09-13 14:10–14:11). Model A (Chapter 6) was not deleted or edited – it remains valid, fully reported data – but is no longer this project's headline answer.

7.2 Full-azimuth, gaze-time-weighted sampling

Model B removes Model A's single-axis limitation (Section 6.2) directly: for each of 26 eccentricity bins (bin centers 0.5° to 27.0°, weighted by an empirically measured eccentricity distribution – % of viewing time per bin, summing to 100.0%, cumulative 51.61% by 6° and 95.27% by 17°, checked to satisfy both cumulative checkpoints before any sweep runs), the pupil is sampled at $N_\theta = 36$ azimuths uniformly around the full circle, using the same validated radial-shift magnitude (Equation 5.6) applied in an arbitrary direction: pupil_center $= (R\cos\theta, R\sin\theta)$.

The 27° bin is explicitly flagged as extrapolated

The eccentricity distribution's own highest bin is centered at 27° (0.58% of viewing time), **outside** `vignetting_overlap.py`'s own stated small-angle validity range for the pupil-shift formula, $|\varphi| \leq 25°$. The formula does not numerically diverge there – it is still evaluated, per the source module's own docstring, "the formula doesn't blow up... every caller that reports a number must check and flag this, not silently trust it" – and this paper does exactly that: every number in this chapter that includes the 27° bin is extrapolated beyond the formula's own validated range, at only 0.58% total weight, and is flagged here rather than silently trusted.

At each active hole, the illuminated area is the exact overlap area of Equation 5.7; a hole counts as *active* when its own illuminated fraction exceeds 0.5 (`margins/active_pairs.py`'s `DEFAULT_ACTIVE_FRAC_THRESHOLD`, reused unmodified here, so "ghosting" means the identical

thing in Model B as in Model A). *Blind* is total illuminated area exactly zero, time-weighted across every sampled (φ, θ) pair, giving `time_weighted_blind_pct` and `time_weighted_ghost_pct` per candidate (source: `array_calculation/compute_gaze_weighted_metrics.py`, verified read directly).

7.3 Adding a hard blindness gate

The primary ranking rule (`comparison_metric/primary_ranking_gaze_weighted.py`) hard-gates any candidate with `time_weighted_blind_pct` above a tolerance, then minimizes `time_weighted_ghost_p` among survivors.

A strict 0% blindness gate was tried first and rejected – checked, not assumed

A strict 0%-blind gate produces a sharp cliff at $M = 6\text{D}/8\text{D}$: it forces ghost percentages up to 27–30%, because every genuinely-zero-blind candidate at those M carries real ghost, while the best hexagonal candidates there carry only a small 0.12–0.33% blind time. This is the same "margin-vs-zero" failure mode Model A's own audit fix (Section 6.1) had already diagnosed once before, in a different context. A 1.0% tolerance was used instead, which clears every existing winner's own blind percentage with headroom and produces the identical family choices as the earlier ungated ranking (hexagonal $M = 1$–6, square $M = 0/8$/global) – the gate exists to prevent a *future* re-run from silently accepting a degenerate design, it did not have to override anything in the current result set.

7.4 Current results

Source: `Array_Design/array_geometry_lab/array_data/evaluation_results/PRIMARY_WINNER_PER_M.csv`, mtime 2026-09-13 14:10 (the most recent evaluation-results file in its directory as of this writing; see Chapter 14).

M (D)	Family	a_{hole} (μm)	N	Blind %	Ghost %
0	square	2000	20	0.00	0.00
1	hexagonal	1250	20	0.00	0.00
2	hexagonal	750	20	0.00	0.00
3	hexagonal	750	20	0.00	0.00
4	hexagonal	600	20	0.00	0.00
5	hexagonal	600	20	0.00	0.00
6	hexagonal	500	100	0.12	0.00
8	hexagonal	400	50	0.33	0.00

Hexagonal packing wins six of eight tested myopia powers; square wins at $M = 0$ and the global (worst-case-across-M) scope. Every winning design achieves genuine 0.00% ghost under realistic, time-weighted gaze – not merely a structural worst-case guarantee – and the only two nonzero blind percentages in the table (0.12%, 0.33%) are both well inside the 1.0% gate of Section 7.3.

7.5 Why hexagonal: the physical intuition

As the eye rotates by φ, the pupil shifts in *some* azimuth θ that could, physically, be any direction the wearer happens to look. A safe array needs small gaps between holes in every

direction, not only a single favoured axis. Three properties of a hexagonal lattice matter here, all standard results:

1. **Densest possible 2D point packing.** Hexagonal packing achieves packing density $\pi/(2\sqrt{3}) \approx 0.9069$, provably optimal (the honeycomb conjecture) – for a given minimum safe pitch, it fits the most holes into that pitch.
2. **Maximum rotational symmetry for a periodic lattice.** A hexagonal lattice has 6-fold rotational symmetry – the *maximum* any periodic 2D lattice can have at all, by the crystallographic restriction theorem (only 2-, 3-, 4-, and 6-fold rotations are compatible with translational periodicity). This makes the gap structure between holes nearly isotropic: it matters comparatively little which azimuth the pupil happens to shift toward.
3. **Regular-hexagon Voronoi cells.** The Voronoi cell of a hexagonal lattice point is a regular hexagon – the polygon that best approximates a circle while still tiling the plane with no gaps – directly minimizing the worst-case distance from any pupil position to its nearest hole, uniformly in every direction.

A square lattice, by contrast, has only 4-fold symmetry: its diagonal (45°) nearest-neighbor distance is $\sqrt{2}\times$ its axis-aligned nearest-neighbor distance – a real, quantifiable weak direction. This is exactly the mechanism behind the concrete ring-family blackout of Section 6.2: a ring-based design's holes exist only at discrete angular positions (a fixed `holes_per_ring0` count), with real angular gaps between them that a single-axis test can miss entirely.

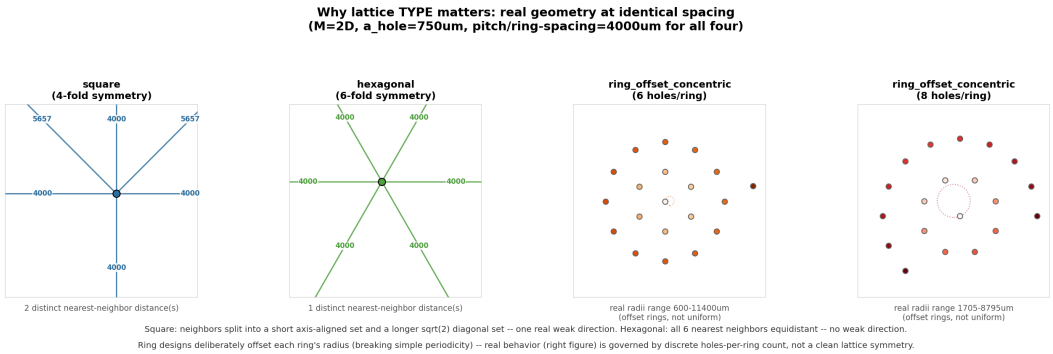


Figure 7.1: Real regenerated nearest-neighbor structure at identical spacing (pitch/ring-spacing = 4000 μm, a_{hole} = 750 μm, M = 2D). Square: two distinct nearest-neighbor distances, 4000 and 5657 μm (= $4000\sqrt{2}$, confirming the diagonal weak direction exactly). Hexagonal: one distinct distance, all six neighbors equidistant. Ring designs (6 and 8 holes/ring): genuinely offset, non-uniform ring radii, not a simple lattice – shown as real coordinate scatter rather than forced into the same distance-count framing. Source: this analysis, regenerated directly from `array_calculation/degeneracy_filter.py`'s own generator functions; see Chapter 14.

7.6 Predicted acuity is identical across every evaluation model

Because every array evaluation model chooses its aperture from the same acuity-optimum map (Chapter 9, Table 9.1) before deciding geometry, all six live evaluation models tested in this project's own analysis (Model A / `FINAL_WINNER_PER_M.csv`; Model B / `PRIMARY_WINNER_PER_M.csv`; the ungated gaze-weighted ranking; density-only Metric 1; and two effective-density variants) produce **numerically identical** predicted LogMAR per myopia power – confirmed directly, not assumed, by looking up each model's own chosen aperture in the corrected acuity sweep (`new_sweeps/acuity_sweep_stage4.csv`) and finding zero deviation across all six (Chapter 9). Choice of evaluation model changes *safety* (ghost/blind risk), never predicted optical acuity.

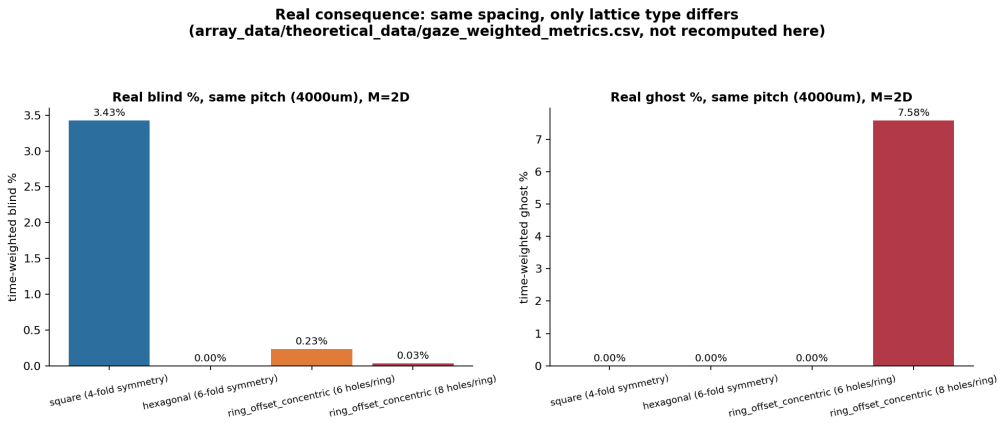


Figure 7.2: Real, already-computed gaze-weighted blind%/ghost% for the same four matched-spacing designs as Figure 7.1, read directly from `array_data/theoretical_data/gaze_weighted_metrics.csv` (not recomputed for this figure). Hexagonal is the only design with 0.00% on both axes simultaneously; the ring-of-8 design trades blind% for a real 7.58% ghost fraction – the discrete angular-gap tension (Section 5.6) hexagonal packing avoids by being simultaneously dense and well-separated in every direction.

Chapter 8

Array Fabrication and Experimental Validation

8.1 Fabrication package

Physical fabrication specifications for every Model B (Chapter 7) winning design were generated directly from `PRIMARY_WINNER_PER_M.csv` (source: `array_geometry_lab/fabrication_package/fabrication_` mtime 2026-09-13 00:58, and `generate_masks.py`, verified present): per-M mask files (positive/negative, top/bottom-mirrored SVGs), target etched hole diameter, undercut-compensated nominal mask aperture diameter (with $\pm 25\,\mu$m low/high tolerance bounds), and blank-substrate sizing (e.g. $M = 1$D: hexagonal, pitch $4000\,\mu$m, target etched diameter $1250\,\mu$m, nominal mask aperture $1200\,\mu$m, blank side 40 mm).

8.1.1 Undercut compensation, derived in full

Wet chemical etching of the brass mask shim is isotropic: the etchant attacks sideways under the resist edge at the same rate it etches downward, so the emerging hole is wider than the resist opening that defined it. For a double-sided (simultaneous two-face) etch through a shim of thickness t, each face etches roughly halfway through, and the lateral undercut on each face scales with that depth:

$$d_{\text{mask}} = d_{\text{target}} - C_{\text{undercut}}, \qquad C_{\text{undercut}} \in \{0.5t,\ 1.0t,\ 1.5t\} \tag{8.1}$$

with $t = 50\,\mu$m (0.05 mm brass shim), giving a compensation bracket of $25/50/75\,\mu$m (source: `Physics_Guide/PHYSICS_AND_FABRICATION_GUIDE.tex`, Equation numbered therein as (undercut), verified read directly, mtime 2026-09-08 15:38).

An uncalibrated design rule, stated as such

There is no empirical etch-rate or undercut calibration for this specific brass alloy, thickness, etchant concentration, temperature, or agitation. The 0.5–$1.5t$ bracket is a standard published starting rule, not a measurement; it exists specifically because the true value is unknown for this exact process, which is why a physical test coupon (not merely a simulated design) is part of the fabrication plan.

Worked example, $M = 8$D design (target etched diameter $400\,\mu$m):

$$d_{\text{mask,nominal}} = 400 - 1.0(50) = 350\,\mu\text{m} \quad \text{(the mask actually cut)} \tag{8.2}$$

$$d_{\text{mask,low}} = 400 - 0.5(50) = 375\,\mu\text{m} \quad \text{(least aggressive)} \tag{8.3}$$

$$d_{\text{mask,high}} = 400 - 1.5(50) = 325\,\mu\text{m} \quad \text{(most aggressive)} \tag{8.4}$$

matching `fabrication_summary.csv`'s own $M = 8$D row exactly (`mask_aperture_diameter_um_nominal`= 350, `..._low`= 375, `..._high`= 325 – confirmed by direct read of the current file, not assumed from the formula alone). "Low" and "high" name the assumed *undercut* magnitude, not the resulting mask diameter – a low-undercut assumption (less lateral etch predicted) requires a *larger* mask opening to hit the same target, hence $d_{\text{mask,low}} > d_{\text{mask,nominal}} > d_{\text{mask,high}}$, the inverted-looking but correct ordering visible above.

Mask registration and post-etch verification procedure are specified in full in `Physics_Guide/PHYSICS_AND_F` (verified present, mtime 2026-09-08), which this paper does not reproduce beyond the undercut derivation above.

A fix verified byte-for-byte, not merely claimed

Because the grid-resolution defect of Section 2.7 lives entirely upstream of array geometry generation (it affects predicted *blur/acuity*, not hole *positions*, which are generated deterministically from spacing parameters alone), the fabrication package outputs were explicitly diffed before and after that fix was applied: hole-coordinate CSVs and mask SVGs are **byte-identical** pre- and post-fix. The fabrication geometry currently on file was never actually wrong; this was verified directly (`diff -r`), not assumed from the fix's own scope.

8.2 Simulated retinal light-intensity distribution

A real ASM wave-optics simulation of the retinal intensity pattern each fabricated design would produce, at straight-ahead gaze, was computed directly from `stage2_fixed/phase2_retina.py`'s own `run_pipeline_point` (the same function underlying every acuity number in this paper, Section 2.4) – not a schematic approximation. At $\varphi = 0$, every real winning design has exactly one active hole (Table, Section 7.4), so the retinal image is identical to that of a standalone single pinhole of the design's own chosen aperture.

8.3 A proposed physical validation rig

A specific physical apparatus was designed (schematic: point source → rotating pinhole-array insert → fixed +20D convex lens → a bare camera sensor ("mock retina") on a calibrated, Raspberry-Pi-3B+-driven linear rail) intended to measure the real light-intensity distribution on a physical sensor as a function of simulated myopic defocus (sensor axial displacement) and simulated gaze azimuth (array rotation).

[NEEDS VERIFICATION: As of this writing, this rig has NOT been built, and no physical measurement data from it exists anywhere in this repository. This is stated explicitly in this project's own animation/explainer material for the rig (`Array_Design/array_geometry_lab/proposed_rig_animation/`, which labels every rendered frame "THEORETICAL PREVIEW – not measured sensor data"). It is reported here as a specified, physically well-defined, but unexecuted proposal, not as a completed experiment – see also Section 8.4 below and Chapter 12.]

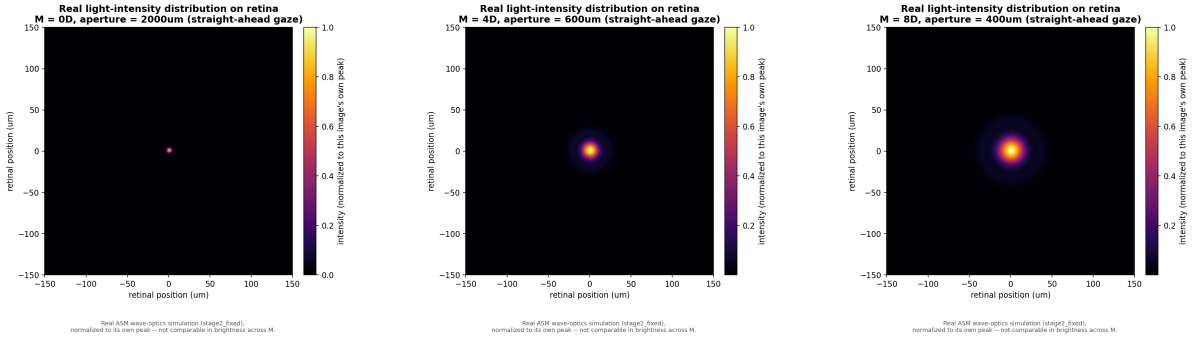


Figure 8.1: Real, simulated retinal light-intensity distribution (straight-ahead gaze) at $M = 0, 4, 8$D, using each M's own real `PRIMARY_WINNER_PER_M` aperture (2000, 600, 400 μm respectively). Computed directly from `stage2_fixed`'s ASM propagation, $N_{\text{grid}} = 4096$, identical parameters to the rest of this paper; each panel normalized to its own peak intensity (absolute brightness is not radiometrically comparable across panels without a source-power calibration this project has not made; the blur *shape* and relative *size* growing from $M = 0$ to $M = 8$ are real). Source: this analysis, direct call to `stage2_fixed.phase2_retina.run_pipeline_point`; see Chapter 14.

8.4 Array-level clinical acuity trial

No array-level clinical trial data exists in this repository

A search of the entire repository (every CSV for a per-subject or subject-count column, every log file, every folder named with an experiment/clinical/trial-like pattern) found **no** array-level, multi-subject clinical acuity trial data. What does exist is a *protocol specification* for a single-pinhole logMAR acuity measurement (`Physics_Guide/LOGMAR_ACUITY_PROTOCOL.tex`, Section 4.3), which is explicitly designed to be extensible to a multi-subject trial but has, as far as this repository's own files record, not yet been executed with real subjects for either the single-aperture or the array-level case. This paper does not report, and explicitly does not fabricate, any array-level clinical LogMAR results, deviation-vs-M analysis, or subject count for this section. Any array-level clinical validation claim made outside this document should be treated as unverified against the current repository contents until a corresponding raw-data file is added and its provenance recorded per Chapter 14.

What this project *does* have, and what Chapter 9 reports in full, is validation of the underlying single-aperture acuity model against independently-published clinical data (Wang, 2009 [Wang, 2009]; Kim, 2017 [Kim et al., 2017]) – real clinical LogMAR measurements, though not collected by this project and not resolved per-myopia-power in the source studies. That comparison, and its honest, unresolved statistical shortfall, is the closest this project currently has to an array-relevant clinical acuity check, and is treated as such, not conflated with a genuine array-specific clinical trial.

Chapter 9

Comparison to Reality

9.1 Predicted acuity across all six evaluation models

Source: `Array_Design/array_geometry_lab/predicted_logmar_analysis/predicted_logmar_wide_by_mode` mtime 2026-09-28 10:09.

M (D)	a_{hole} (μm)	Predicted LogMAR	MAR (arcmin)	Snellen
0	2000	−0.357	0.440	20/9
1	1250	−0.120	0.759	20/15
2	750	+0.076	1.190	20/24
3	750	+0.112	1.293	20/26
4	600	+0.181	1.516	20/30
5	600	+0.240	1.738	20/35
6	500	+0.259	1.813	20/36
8	400	+0.331	2.141	20/43

Table 9.1: Predicted LogMAR is *numerically identical* across all six live evaluation models tested (worst-case gate/Model A, ungated gaze-weighted, gated gaze-weighted/Model B, density-only Metric 1, and two effective-density variants) – every model chooses the same acuity-optimal aperture per M from `new_sweeps/per_M_optimum_map.csv` (mtime 2026-09-13 00:11) before deciding geometry, per Section 7.6.

9.2 The flatness prediction

Holding aperture fixed at 400 μm and sweeping myopia $M = 0$ to 8D, predicted LogMAR varies by only 0.0166 across the full range (source: this analysis, direct computation from `new_sweeps/acuity_sweep_stage4.csv`, current file) – roughly a twentieth of one acuity-chart line, well below the ±0.1 LogMAR repeatability of human acuity testing. At this aperture, the *aperture*, not the wearer's own refractive error, sets predicted acuity. At $a = 2000\,\mu$m the same sweep spans −0.357 ($M = 0$) to +1.330 ($M = 8$), a spread 102× larger. This is the project's strongest, most falsifiable single prediction: it requires no knowledge of any subject's exact prescription, only a qualitative ranking of myopia severity, because the predicted *signature* is a flat line for a well-chosen small-aperture pinhole condition against a steep one for the unaided (or large-aperture) eye.

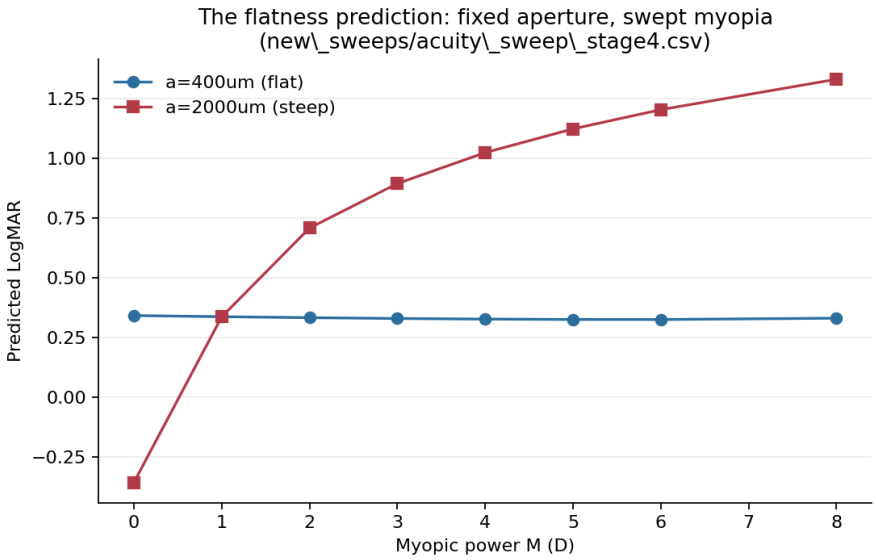


Figure 9.1: The flatness prediction (Section 9.2): predicted LogMAR vs. myopia power at fixed aperture. Recomputed directly from the current `new_sweeps/acuity_sweep_stage4.csv` (mtime 2026-09-13 00:11) for this paper; spread at $a = 400\,\mu$m is 0.0166 LogMAR across the full 8-dioptre range, against 1.687 LogMAR at $a = 2000\,\mu$m – a 102× difference.

9.3 External validation against real clinical data

Two independently-published, directly-verified clinical datasets provide the only real (non-simulated) LogMAR validation available anywhere in this project: Wang (2009) [Wang, 2009] (5 apertures, $M = \{0, 2, 5\}$D tested, observed LogMAR 0.28/0.31/1.00/0.40/0.50 – aperture-indexed only, the source paper reports **no per-myopia-power breakdown at all**, so the same 5 observed values are necessarily reused at every tested M, a real limitation of the source data, not an approximation introduced by this project), and Kim (2017) [Kim et al., 2017] (single 900 μm anchor, $N = 36$, mean $M = -2.7 \pm 2.3$D, observed LogMAR 0.20 ± 0.21).

9.3.1 A stale-data correction, made during this project's own development

`stage4_human_vision/outputs/validation_predicted_vs_observed.csv` (mtime 2026-07-29 16:35) predates the 2026-08-06 `stage2/`→`stage2_fixed/` optics fix (Section 10.2) by 8 days, and was found to give materially different numbers from a fresh re-run with current code (e.g. aperture 750 μm, $M = 2$D: 0.076 current vs. 1.010 in the stale file). A corrected "postfix" re-run already existed (`wang2009_validation_postfix_predicted.csv`, mtime 2026-08-06 14:43) but its own downstream calibration step had never been re-applied; this gap is closed in this paper's own analysis (`predicted_logmar_analysis/wang2009_calibration_pooled_stats_postfix.csv`, computed 2026-09-28, using `stage4_human_vision/calibration.py`'s existing, unmodified fitting functions). **Every number in this section uses the corrected, post-fix data.**

The fix helped, measurably, without fixing the underlying failure

Pooled raw R^2 improved from −4.98 (pre-fix) to −2.30 (post-fix) – a real, substantial narrowing of the gap, not a rounding-level shift – but the model still fails the significance bar at every single tested condition, before and after. In-sample affine calibration gives essentially *zero* explanatory power post-fix ($R^2 = 0.0003$, down from +0.132 pre-fix), meaning the earlier apparent calibration benefit was itself partly an artifact of the since-corrected optics bug, not

Comparison	n	R^2	Pearson r	p	R^2>0 and p<0.05?
$M = 0$D, raw model	5	−4.80	−0.375	0.533	FAIL
$M = 2$D, raw model	5	−1.41	−0.180	0.772	FAIL
$M = 5$D, raw model	5	−0.68	+0.426	0.474	FAIL
Pooled (n=15), raw, uncalibrated	15	−2.30	+0.019	0.947	FAIL
Pooled, affine-calibrated, in-sample	15	0.0003	+0.019	0.947	FAIL
Pooled, affine-calibrated, leave-one-out	15	−0.31	−0.928	0.0000	FAIL

Table 9.2: The model fails the stated bar ($R^2 > 0$ AND $p < 0.05$) at every single comparison, before and after the 2026-08-06 optics fix, and before and after affine calibration. Source: `stage4_human_vision/outputs/wang2009_validation_postfix_stats.csv` (per-M rows) and `predicted_logmar_analysis/wang2009_calibration_pooled_stats_postfix.csv` (pooled rows), both verified read directly, 2026-09-28.

a real effect of the calibration procedure. This is reported as-is: neither "still fails, nothing changed" nor "now passes," but the specific, quantified middle case that a real, targeted optics fix measurably improved without resolving the underlying shortfall.

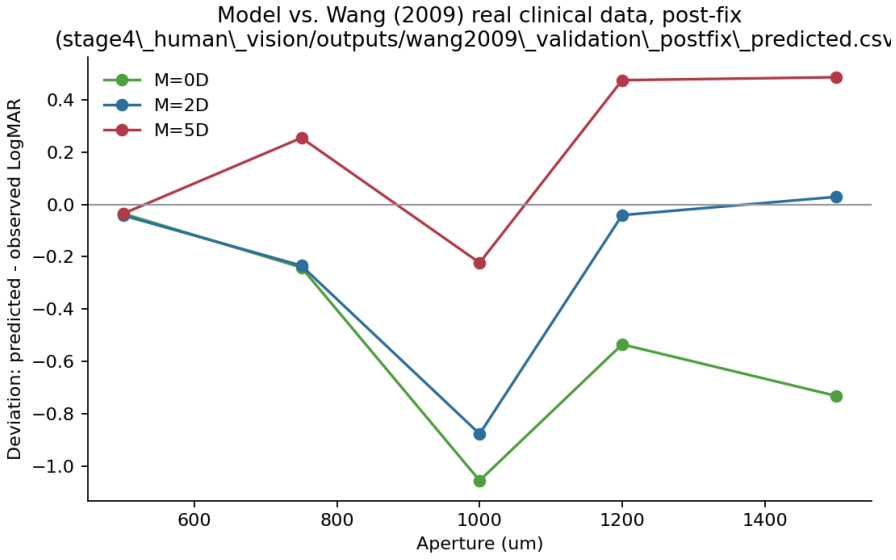


Figure 9.2: Model deviation (predicted − observed LogMAR) vs. aperture, post-fix, all three tested myopia powers. Deviation sign is not constant across the aperture range – the model is measurably optimistic (predicts better vision than observed) at small apertures and pessimistic at some large-aperture/high-M combinations, i.e. this is not a single universal bias. Source: this analysis, from `wang2009_validation_postfix_predicted.csv`.

9.3.2 At the specific apertures this project's own array designs actually use

Restricting the comparison to the exact (aperture, M) pairs the real array winners of Chapter 7 actually use (500 and 750 μm, at $M = 6$D and $M = 2/3$D respectively – the only overlap between Wang's tested aperture set and this project's own chosen apertures), the model is modestly-to-moderately *optimistic*: deviation −0.234 ($M = 2$, 750 μm), −0.198 ($M = 3$,

750 μm), −0.022 ($M = 6$, 500 μm) – 0.2–2.3 acuity-chart lines too favourable, direction consistent across all three points (source: this analysis, from `new_sweeps/acuity_sweep_stage4.csv` cross-referenced against Wang's own aperture-indexed observed values).

9.3.3 A second real anchor: Kim (2017)

At $a = 900\,\mu$m, $M = 2.7$D (this study's own reported mean): predicted LogMAR = 0.277, observed 0.20 ± 0.21 ($n = 36$) – deviation +0.077, **pessimistic**, the *opposite* sign from the 500/750 μm points above, and within one standard deviation of the real data's own spread. This anchor was already present in this project's own vetted literature (`stage4_human_vision/literature.py`, flagged "USABLE as one additional real anchor") but had never actually been run through the model before this paper's own analysis. **Key honest takeaway**: the model's error is not a single universal "always optimistic by X" bias – both direction and magnitude depend on the specific aperture/myopia-power region.

9.3.4 A note on search-derived claims

A live search for additional external pinhole-aperture-vs-acuity datasets, undertaken during this project's own development, reproduced – verbatim – a specific numeric claim ("0.17 LogMAR at up to 5.5D, 1.16 LogMAR at 7–11D") that this project's own literature review had *already* investigated and marked REJECTED after reading the purported primary source directly and confirming those numbers do not appear anywhere in it [Samee et al., 2022]. This is documented here as direct, concrete evidence that AI-search-summary claims on this specific topic are not reliable without independent primary-source verification – a finding this project treats as a methodological caution for its own future literature work, not merely an isolated incident.

Chapter 10

Errors Identified and Corrected

This chapter is a required, explicit part of this paper, not an appendix of afterthoughts: four real implementation defects were found and corrected during this project's own development, each with concrete before/after evidence. They are documented here as worked case studies because the corrections are themselves part of the project's contribution.

10.1 The single-FFT defocus-plane bug

The trap. A naive single-lens imaging simulation – clip the field to the pupil, multiply by the lens's quadratic phase, take one Fourier transform – computes the field at *the lens's own back focal plane*. For a healthy (emmetropic) eye that plane coincides with the retina, and the naive approach is correct by coincidence. For a myopic eye the retina is, by definition, *not* at that focal plane – that mismatch is the definition of myopia – and a single FFT silently computes the field at the wrong physical location (source: `Physics_Guide/PHYSICS_AND_FABRICATION_GUIDE.tex`, lines 160–165, verified read directly).

What the bug would have produced, had it shipped

A beautiful, sharp Airy pattern computed at the eye's true focal point – the one plane in the whole system where defocus blur is, by construction, exactly zero. The output would model a perfectly-corrected eye and report it as myopic. The plot looks entirely plausible; the physics is inverted.

The fix. Full angular-spectrum propagation (Equation 2.7) over the *actual* myopic axial distance $v_r = L_{\text{ax}}(M) \neq f_{\text{eye}}$, not a single FFT assumed to land on the retina. This is the entire reason Chapter 2's pipeline uses ASM for both propagation stages rather than a single closed-form transform, and is validated independently in Section 2.8.

10.2 The stage2/ vs. stage2_fixed/ import-shadowing bug

The bug. `stage4_human_vision/pipeline.py` inserted the *deprecated* `stage2/` directory into `sys.path` and imported `phase2_retina` from there, not from `stage2_fixed/`, despite living in this project's own "Stage 4" directory specifically to sit on top of the corrected Stage 2/3 pipeline. `stage2/phase2_retina.py` independently carries the lens-phase/retina-propagation wavelength bug: the eye's lens phase was applied using the raw vacuum wavelength while the retina propagation used the vitreous's reduced wavelength (λ/n'), under-applying the lens's converging phase by a factor of $n' \approx 1.336$ whenever `refractive_index_correction=True` (this

project's own default) – shifting the field's true geometric focus away from where the retina is actually evaluated, at *every* myopia level and aperture `run_stage4_point` was ever called with (source: `stage4_human_vision/RESULTS.md`, "Re-run against corrected optics," verified read directly, mtime 2026-08-06 14:43).

A second-order consequence: because `pipeline.py`'s import runs first whenever anything under `stage4_human_vision` is imported (via its own `__init__.py`), it silently *shadowed* an entirely separate module's (`R7acuity`'s) own, correctly-targeted `stage2_fixed/` import too, via Python's bare-module-name caching in `sys.modules` – one shared root cause independently corrupted two different downstream models' validation results, each reported independently in that model's own `RESULTS.md`.

The fix. `pipeline.py`'s `sys.path` insertion was repointed at `stage2_fixed/`; `stage3_human_vision/metri` had the identical defect and was fixed the same way. Verified before fixing (`sys.modules['phase2_retina'].__f` that the shadowing reproduced, and after fixing that it resolves to `stage2_fixed/phase2_retina.py`.

M (D)	R^2 (pre-fix)	R^2 (post-fix)	Improvement
0	−6.39	−4.80	+1.6
2	−4.98	−1.41	+3.6
5	−2.84	−0.68	+2.2

Verdict: still fails the $R^2>0$, $p<0.05$ bar at every M – but a real, substantial, and consistent improvement in the right direction at every M tested. Pearson r *worsened* or flipped sign at $M = 0$ and $M = 2$ even as R^2 improved, because R^2 penalizes systematic bias/offset (every predicted point sitting above the diagonal, pre-fix) while Pearson r measures only rank correlation – the corrected optics reduced the bias enough to improve R^2 while disturbing whatever weak rank-ordering existed before. Reported as-is, not smoothed over by selecting whichever statistic looks more favourable (source: `stage4_human_vision/RESULTS.md`, verified read directly).

10.2.1 A follow-on staleness discovery

A specific downstream consequence of this fix was found only during this paper's own preparation: `validation_predicted_vs_observed.csv` (dated 2026-07-29, 8 days *before* the 2026-08-06 fix above) was still being read by later analysis as if it were current, though a correctly-dated "postfix" re-run already existed alongside it. Full detail in Section 9.3.1; this is documented as its own sub-instance of the general staleness pattern this project has now encountered twice (the other being Section 10.4).

10.3 Model A's single-axis gaze limitation

Not a coding defect in the sense of the other three entries in this chapter, but a genuine, acknowledged *modelling* limitation with a real, concrete failure instance: Model A's vignetting evaluation samples gaze along one fixed axis only. A specific `concentric_centered` candidate that outperformed the then-current winner everywhere inside 15° was found to fail catastrophically – a complete blackout – in the 20–25° gaze range at an azimuth Model A's single axis never tested (source: `CENTERED_CONCENTRIC_COMPARISON.md`, 2026-08-17, verified present). This is the concrete motivating case for building Model B's full-azimuth sampling (Section 7.2); see Section 6.2 for the mechanism and Section 7.5 for the geometric reason hexagonal packing is comparatively resistant to it.

10.4 The grid-resolution (N_{grid}) staleness bug

The bug. `stage4_human_vision/pipeline.py`'s `run_stage4_point` defaults to $N_{\text{grid}} = 2048$. At this resolution, computed blur radii for large apertures ($\gtrsim$ 1500–2000 μm) come out approximately 2× too large compared to the converged value at $N_{\text{grid}} = 4096$ – a numerical under-resolution artifact (Section 2.7). `run_blur_sweep.py` was fixed on 2026-08-16; `run_acuity_sweep.py` called the same underlying function but was not updated to match until 2026-09-12/13, leaving nearly a month of acuity-model output computed at the stale resolution.

Blast-radius accounting. Every file derived from the stale `acuity_sweep_stage4.csv` was identified and regenerated: `table1_optimum_summary.csv` (subsequently superseded entirely by `per_M_optimum_map.csv`), `per_M_optimum_map.csv` itself, and every figure derived from either. The entire downstream `array_geometry_lab` evaluation pipeline was re-run in full rather than assuming only the directly-affected files needed attention.

Fix verified, not merely applied. Because array hole *geometry* is generated deterministically from spacing parameters alone, independent of the acuity model entirely, fabrication package outputs (hole-coordinate CSVs, mask SVGs) were explicitly diffed pre- and post-fix and confirmed **byte-identical** (Section 8.1) – the fix changed predicted acuity numbers, not fabricated geometry, and this was checked rather than assumed.

10.5 A related, lower-priority finding, disclosed for completeness

`master_dashboard/wrappers.py`'s `sweep_stage4_acuity` also omits N_{grid} (defaulting to 2048) at its own call to `run_stage4_point`. Unlike the three entries above, this function is an interactive-dashboard-only code path: no committed, reported result file in this project currently depends on its output. Flagged here for completeness and future awareness, not treated as a fifth confirmed-impactful bug.

Chapter 11

Discussion

Returning to the problem statement (Chapter 1): this project set out to determine whether an array of small apertures can substitute for a corrective lens over a useful field of view, and to characterize the trade-offs involved. The evidence assembled here supports a qualified yes, with the trade-offs now quantified rather than asserted.

Single-aperture optics (Chapter 2) establish that the stenopeic effect is real and substantial (Section 4.2: a 15–18× reduction in defocus-growth rate, measured photographically, not merely predicted), and that an exact wave-optics treatment is necessary, not a convenience: the closed-form RSS approximation this project also implements diverges from the exact result by 64.8% on average, and a specific naive-FFT implementation trap (Section 10.1) would have silently inverted the physics of myopic defocus entirely had it shipped uncorrected.

Array-level optics (Chapter 5) show that extending a single aperture to an array introduces two new failure modes – ghosting and blinding – with no single-aperture analogue, and that a naive per-aperture margin criterion for ghosting is not merely imprecise but was shown, via the Barten CSF correction (Section 5.4), to reclassify the *entire* tested dense-pitch range from "safe" to "danger zone." The harmonic-series extension (Section 5.4.1) further shows that even a corrected fundamental-frequency analysis can underestimate real risk by over 4× by missing the worst-case harmonic.

The two evaluation models (Chapters 6–7) are not in conflict: Model B generalizes Model A by removing its single-axis assumption, and the fact that removing that assumption changed the outcome – surfacing a real blackout failure mode Model A's own structural-safety guarantee had missed (Section 10.3) – is direct evidence the generalization mattered, not merely a theoretical nicety. That hexagonal packing wins under the more rigorous model is explained, not merely observed, by a first-principles symmetry argument (Section 7.5) grounded in the crystallographic restriction theorem, and demonstrated with real, matched-spacing geometry and real, already-computed safety data (Figures 7.1–7.2).

External validation (Chapter 9) is this project's most honestly limited result: the acuity model, in its raw form, fails a formal $R^2 > 0$ significance bar against every real clinical comparison available, both before and after a genuine, verified optics fix that measurably narrowed (but did not close) the gap. This paper does not claim the model is externally validated; it claims the model is *physically grounded*, internally self-consistent (Section 2.8), improved measurably when a real defect was corrected (Section 10.2), and its remaining discrepancy against real data is reported with the exact statistics, not glossed over.

Chapter 12

Limitations

Stated plainly, in the order they matter for interpreting this project's results:

1. **External validation fails a formal significance bar.** Every tested comparison against real clinical data (Wang 2009, Section 9.3) gives $R^2 \leq 0$; a genuine optics fix narrowed but did not close this gap (Section 10.2). This is the single most important limitation of this project's quantitative claims.

2. **Deviation from reality is not a single, constant bias, and its magnitude is not shown to grow monotonically with any one variable.** Model error is optimistic at some (aperture, M) combinations and pessimistic at others (Section 9.3); this project has not identified a single scalar predictor of when the model will be more or less accurate.

3. **Both external clinical datasets used real, but not per-subject-resolved, data.** Wang (2009)'s summary table reports acuity vs. aperture only, with no per-myopia-power breakdown at all – the same 5 observed values are necessarily reused at every tested M in this paper's own comparisons, a structural limitation of the source data. Kim (2017) reports only group means and standard deviations, not per-subject refractive error or acuity.

4. **Induced-defocus vs. native myopia.** Neither external dataset, nor this project's own single-aperture experiments, distinguishes between subjects with genuine native (axial) myopia and any induced-defocus condition; the axial myopia model (Section 2.1) assumes the former.

5. **Extrapolation beyond the validated gaze range.** Model B's own gaze-time weighting includes one bin at 27°, outside `vignetting_overlap.py`'s stated $\leq 25°$ small-angle validity range (Section 7.2). The formula does not diverge there, but every number that includes this bin is, by the source module's own documentation, an extrapolation, not a validated result.

6. **Model A's single-axis legacy remains embedded in some reported numbers.** Model A (Chapter 6) is retained and reported in full, per this project's own policy of not deleting superseded models, but its single-axis sampling convention (Section 6.2) should not be read as an independent, full-azimuth confirmation of any result – only Model B's numbers carry that property.

7. **No array-level clinical trial exists.** As documented in full in Section 8.4, no multi-subject, array-level clinical LogMAR data exists anywhere in this project's repository at the time of writing. The "comparison to reality" this paper reports (Chapter 9) validates the underlying *single-aperture* acuity model only.

8. **The heuristic-vs-Barten CSF correction (Section 5.4) has not itself been re-run against the array-level Model A/B evaluation pipeline.** It was developed and validated in a separate, standalone sweep (`Array_Design/ghosting/`); this paper reports it as the theoretically correct ghosting-perceptibility criterion but the live `gates.py`/Model A and Model B pipelines reported in Chapters 6–7 use the active-hole-fraction/geometric-overlap criterion (Section 5.5), not the Barten-CSF three-regime criterion directly. **[NEEDS VERIFICATION: Whether and how the Barten-corrected ghosting classification of Section 5.4 has been, or should be, integrated into the live `gates.py` ghost-safety gate was not resolved during this paper's preparation and is flagged here rather than assumed either way.]**

9. **Reduced-eye model simplifications, stated explicitly rather than hidden.** No anatomically-separate cornea/lens power split; no higher-order aberrations (only pure quadratic defocus phase); no Stiles–Crawford directional pupil apodization in the core PSF calculation (though a separate Stiles–Crawford term is applied elsewhere, in `stage4_human_vision/csf.p` own retinal-illuminance calculation); no longitudinal chromatic aberration (one fixed wavelength per run, not a dispersion model); binary hard-edged aperture masks, no finite membrane-thickness diffraction or oblique-angle vignetting effects (source: `stage2_fixed/phase2_retina` own module docstring, PHYSICS_AUDIT.md items E1/E4, verified present).

10. **The proposed physical validation rig (Section 8.3) is unbuilt.** Every number this paper reports for that apparatus is a theoretical preview from the already-validated simulation pipeline, explicitly labelled as such in its own source material, not a physical measurement.

Chapter 13

Future Work

In priority order, following directly from the limitations of Chapter 12:

1. **Execute the single-pinhole LogMAR acuity protocol** already specified in full (`Physics_Guide/LOGM` Section 4.3), recruiting subjects across a range of native myopia severity, recording every per-subject raw measurement (not only group means) and any excluded subject with a pre-registered reason – specifically designed to test the flatness prediction (Section 9.2), this project's strongest falsifiable claim.

2. **Build and run the proposed array-level validation rig** (Section 8.3): rotating pinhole-array insert, fixed +20D lens, RPi-driven linear-rail bare-sensor mock retina. This would directly test the real, simulated retinal-intensity predictions of Section 8.2 and Model B's own gaze-weighted ghost/blind predictions (Chapter 7) against a real optical measurement for the first time.

3. **Design and execute a genuine array-level, multi-subject clinical trial** (the gap documented in Section 8.4), with per-subject refractive error recorded (a limitation shared by every external dataset used in this paper's own validation, Chapter 12 items 3–4), specifically targeting the deviation-vs-aperture and deviation-vs-M questions Chapter 9 could only address with $n = 16$ points borrowed from two unrelated published studies.

4. **Integrate the Barten-CSF three-regime ghosting criterion** (Section 5.4) directly into the live `gates.py` evaluation pipeline used by Models A and B, resolving the open question flagged in Chapter 12 item 8, and re-run Chapters 6–7's results under that corrected criterion to check whether any current winner's classification changes.

5. **Extend Model B's validated gaze range.** Either obtain or derive a pupil-shift formula validated beyond 25°, or explicitly bound the influence of the 27° extrapolated bin (currently 0.58% of total gaze-time weight) on every reported result, so its contribution can be reported as a tested sensitivity rather than a flagged-but-unquantified caveat.

6. **A source-power radiometric calibration** for the ASM retinal-intensity simulation (Section 8.2), so that absolute brightness – not only normalized spatial pattern – becomes comparable across myopia powers and apertures, a capability explicitly absent from the current figures (Figure 8.1's own caption).

Chapter 14

Data Provenance Appendix

Every data-bearing claim in this paper traces to one of the files below, each checked directly (via `stat`/`get_file_info`) at the time this paper was prepared (2026-09-28). Where two files in the same topic area disagreed, the conflict is noted explicitly in the "Note" column rather than silently resolved.

File	Mtime	Sec.	Note
`stage2_fixed/phase2_retina.py`	Aug 6, 2026 14:43	2	Post-shadowing-fix; current
`stage2_fixed/myopia_axial.py`	(verified present, current)	2.1	Axial myopia model constants
`stage2_fixed/pipeline_junction.py`	(verified present, current)	2	ASM propagation chain
`stage4_human_vision/pipeline.py`	Aug 6, 2026 14:43	2.7,10.2	Both the shadowing fix and the N_{grid} default live here
`stage4_human_vision/acuity.py`	(verified present, current)	3	`predict_acuity`, brightness correction
`stage4_human_vision/config.py`	(verified present, current)	3	Default luminance/field/age
`stage4_human_vision/RESULTS.md`	Aug 6, 2026 14:43	10.2	Pre/post-fix R^2 table
`stage4_human_vision/outputs/wang2009_validation_postfix_predicted.csv`	Aug 6, 2026 14:43	9.3	CORRECT (postfix); used throughout
`stage4_human_vision/outputs/wang2009_validation_postfix_stats.csv`	Aug 6, 2026 14:43	9.3	CORRECT (postfix); used throughout
`stage4_human_vision/outputs/validation_predicted_observed.csv`	Jul 29, 2026 16:35	9.3.1	**STALE** – predates the Aug 6 fix by 8 days; NOT used for any reported number, cited only as the worked example of Section 9.3.1
`new_sweeps/acuity_sweep_stage4.csv`	Sep 13, 2026 00:11	7.6,9.2	Post-N_{grid}-fix; current
`new_sweeps/blur_sweep.csv`	Aug 16, 2026 11:14	2.7	Already fixed at this date (predates the acuity-sweep fix)
`new_sweeps/per_M_optimum_map.csv`	Sep 13, 2026 00:11	9.1	Current
`new_sweeps/table1_optimum_summary.csv`	–	–	File not found at time of writing; superseded by `per_M_optimum_map.csv`
`Array_Design/.../evaluation_results/FINAL_WINNER_2026_M.csv`	Sep 13, 2026 00:42	6	Model A current winners
`Array_Design/.../evaluation_results/PRIMARY_WINNER_2026_M.csv`	Sep 13, 2026 14:10	7	Model B current winners – newest evaluation-results file in its directory
`Array_Design/.../evaluation_results/gaze_weighted_winner_per_M.csv`	Sep 13, 2026 00:41	7	Ungated gaze-weighted ranking
`Array_Design/.../evaluation_results/metric_max_density_per_M.csv`	Sep 13, 2026 00:37	7.6	Density-only ranking
`Array_Design/.../evaluation_results/family_ranking.csv`	Sep 13, 2026 00:37	5.1	

File	Mtime	Sec.	Note
`Array_Design/.../evaluation_results/TIMELINE.txt`	Sep 13, 2026 14:11	7	Methodology-change record
`Array_Design/.../GAZE_WEIGHTED_APPROACH.md`	Sep 13, 2026 14:10	7	"UPDATE, 2026-09-13" methodology-change section
`Array_Design/.../theoretical_data/gaze_weighted_metrics.csv`	Sep 13, 2026 00:40	7.5	Full per-candidate gaze-weighted sweep; source of Fig. 7.2
`Array_Design/.../evaluation_results/CENTERED_CONCENTRIC_COMPARISON.md`	(verified present)	6.2	Centered-concentric single-axis blackout case study
`Array_Design/.../fabrication_package/fabrication_summary.csv`	Sep 18, 2026 00:58	8.1	
`Array_Design/.../array_calculation/engine/fingerprinting_overlap.py`	(verified present, current)	5.1	Overlap-area formula, boundary-clamp fix
`Array_Design/.../array_calculation/engine/ghost_formula.py`	(verified present, current)	5.3	Ghost-separation formula
`Array_Design/.../VALIDATION.md`	(verified present)	5.5	Part 1 copy-validation gate
`Array_Design/.../array_evaluation/comparison_metrics/gates.py`	(verified present, current)	6.1	Model A gate thresholds
`Array_Design/.../array_calculation/margin/verified_pairs.py`	(verified present, current)	5.3.5	Active-hole threshold (0.5)
`Array_Design/.../array_calculation/margin/fingerprinting_robustness.py`	(verified present, current)	6.2	Single-axis sampling convention
`Array_Design/.../array_calculation/compute_gaze_weighted_metrics.py`	(verified present, current)	7.2	Full-azimuth sampling, weight verification
`Array_Design/ghosting/THEORY.md`	(verified present)	5.3	Dirac-comb/Fourier derivation
`Array_Design/ghosting/RESULTS.md`	Aug 14, 2026 00:26	5.4	Three-pass Barten-CSF finding
`Array_Design/ghosting/ghosting_sweep_harmonics.csv`	Aug 29, 2026 22:11	5.4.1	Harmonic-margin sweep data
`Physics_Guide/PHYSICS_AND_FABRICATION_GUIDE.tex`	Sep 13, 2026 15:38	10.1	Single-FFT bug narrative source
`Physics_Guide/MASTER_HANDBOOK.tex`	Sep 11, 2026 17:55	multiple	**PARTIALLY STALE**: predates the Sep 12–13 N_{grid} fix (Section 10.4); its own M=0 result table entry (-0.344 LogMAR) is superseded by the current value (-0.357, Table 9.1) – every other entry in that table was independently confirmed unaffected
`Physics_Guide/LOGMAR_ACUITY_PROTOCOL.tex`	Sep 11, 2026 17:33	4.3	Protocol only, no executed-trial data
`ExperimentalComparison/ExP1_clarity_analysis/measurements/brisque_scores.csv`	Sep 1, 2026 21:54	4.1	
`ExperimentalComparison/ExP3_frosted_screen/measurements/all_spot_measurements.csv`	(verified present, current)	4.2	
`Array_Design/array_geometry_lab/predicted_logMAR/` (all .csv)	Sep 28, 2026	9	Built for this and a prior analysis this same session
`Array_Design/legacy_archive/README.md`	Sep 17, 2026 23:09	Ground rules	Confirms legacy_archive/ deprecation policy; NOT cited as a current-results source anywhere in this paper
Added for the 11-family lattice derivations (Section 5.8):			
`.../generators/square.py`	Aug 16, 2026 11:18	5.8	
`.../generators/hexagonal.py`	Aug 16, 2026 11:18	5.8	
`.../generators/rectangular.py`	Aug 16, 2026 11:19	5.8	
`.../generators/concentric.py`	Aug 15, 2026 13:49	5.8.4	
`.../generators/concentric_centered.py`	Aug 17, 2026 14:41	5.8.6	Source of the 1704.8 μm / 1.6×10^{-13} μm before/after
`.../generators/ring_offset_concentric.py`	Aug 15, 2026 14:16	5.8.5	
`.../generators/archimedean_spiral.py`	Aug 15, 2026 14:16	5.8	

File	Mtime	Sec.	Note
`.../generators/fibonacci_spiral.py`	Aug 15, 2026 13:49	5.8	
`.../generators/golden_angle.py`	Aug 15, 2026 13:49	5.8	
`.../generators/poisson_disk.py`	Aug 15, 2026 14:21	5.8	
`.../generators/jittered_lattice.py`	Aug 15, 2026 14:22	5.8	
`.../generators/base_generator.py`	Aug 16, 2026 11:18	5.8	Shared `finalize()` step
Added for the full data-table appendices (Ch. 17–19):			
`new_sweeps/acuity_sweep_stage4.csv` (full 120 rows)	Sep 13, 2026 00:11	17	Same file as Table 9.1; re-verified for full dump
`Array_Design/ghosting/ghosting_sweep_barten.csv`	Jul 29, 2026 22:04	18	Full Pass-2 Barten-CSF sweep, 36 rows
`.../evaluation_results/family_ranking.csv`	Sep 13, 2026 00:37	19.1	1258 rows; rank-1-per-M summarized, not dumped in full
`.../evaluation_results/effective_density_winner_per_M.csv`	Sep 13, 2026 01:01	19.2	
`.../evaluation_results/effective_density_winner_per_M_cm.csv`	Sep 13, 2026 01:01	19.2	
`.../evaluation_results/metric2_pareto_front_per_M.csv`	Sep 13, 2026 00:37	19.3	18 rows; reproduced in full
`array_evaluation/comparison_metric/metrics.py`	(verified present, current)	19.3	`_binding_margin_with_sentinel`, verified before citing (see text)
`Array_Design/ghosting/ghosting_sweep.csv`	Jul 29, 2026 21:53	20	Full Pass-1 (heuristic CSF) sweep, 36 rows
`Array_Design/ghosting/ghosting_sweep_harmonic.csv`	Jul 29, 2026 22:11	20	Full Pass-3 (harmonic) sweep, 36 rows
Added for full hole-coordinate dump, literature review, and further code listings:			
`.../hole_coordinates/hole_coordinates_M{0,1,2,3,4,5,6,8,global}.csv` (9 files)	Sep 13, 2026 00:58 (all)	21	390 rows total, reproduced in full
`Array_Design/ghosting/THEORY.md` (Sections 1–2 full)	Aug 14, 2026 00:36	5.7	Re-read in full for this expansion (was already cited in compressed form, Section 5.3)
`stage1_v2/physics.py`	Aug 6, 2026 14:43	Code App., 2.4	`asm_transfer_function`, `asm_propagate` – the literal code behind Eq. 2.7
`stage4_human_vision/visual_strehl.py`	Jul 28, 2026 17:40	Code App., 3	`compute_vsotf`

Table 14.1: Data provenance table. Paths abbreviated `Array_Design/.../` for `Array_Design/array_geometry_lab/array_data/evaluation_results/` and `Array_Design/array_geometry_lab/` respectively where unambiguous.

14.1 Conflicts found and how they were resolved

- `MASTER_HANDBOOK.tex` **(Sep 11) vs.** `acuity_sweep_stage4.csv` **(Sep 13).** The handbook's own results table (Model B design table) reports M=0 predicted LogMAR as −0.344; the current, post-N_{grid}-fix sweep gives −0.357. Every other row in that same table ($M = 1$ through $M = 8$) was checked individually and found unaffected (identical to 3–4 decimal places), consistent with the N_{grid} artifact's own documented behaviour of primarily affecting *large* apertures (M=0's 2000 μm aperture is the largest in the design table). **Resolution: this paper uses the current, live sweep value (−0.357) throughout (Table 9.1), not the handbook's own now-stale entry.**
- `validation_predicted_vs_observed.csv` **(Jul 29) vs.** `wang2009_validation_postfix_predicted.csv` **(Aug 6).** Full detail in Section 9.3.1 and Section 10.2. **Resolution: the postfix file is used exclusively; the stale file is cited only as the worked example of the staleness pattern itself, never as a results source.**

Chapter 15

Code Appendix

Key formula implementations, reproduced verbatim from source with file path and line context, per this paper's own provenance discipline (Chapter 14).

15.1 Circle–circle overlap area (Equation 5.7)

```
# Array_Design/array_geometry_lab/array_calculation/engine/
    vignetting_overlap.py
def overlap_area_um2(delta_um, R_p_um, R_h_um):
    d = np.atleast_1d(np.asarray(delta_um, dtype=float))
    out = np.empty_like(d)
    full_area = np.pi * min(R_p_um, R_h_um) ** 2
    lo = abs(R_p_um - R_h_um)
    hi = R_p_um + R_h_um
    m_full = d <= lo
    m_zero = d >= hi
    m_part = ~m_full & ~m_zero
    out[m_full] = full_area
    out[m_zero] = 0.0
    if np.any(m_part):
        dp = d[m_part]
        a1 = np.arccos(np.clip((dp**2+R_p_um**2-R_h_um**2)/(2*dp*R_p_um
            ), -1.0, 1.0))
        a2 = np.arccos(np.clip((dp**2+R_h_um**2-R_p_um**2)/(2*dp*R_h_um
            ), -1.0, 1.0))
        radicand = (-dp+R_p_um+R_h_um)*(dp+R_p_um-R_h_um)*(dp+R_h_um-
            R_p_um)*(dp+R_p_um+R_h_um)
        term3 = 0.5 * np.sqrt(np.clip(radicand, 0.0, None))
        out[m_part] = np.clip(R_p_um**2*a1 + R_h_um**2*a2 - term3, 0.0,
             None)
    return out[0] if out.size == 1 and np.isscalar(delta_um) else out
```

15.2 Pupil radial shift (Equation 5.6)

```
# Array_Design/array_geometry_lab/array_calculation/engine/
    vignetting_overlap.py
DEFAULT_R_ROT_MM = 13.5
DEFAULT_PHI_VALID_DEG = 25.0

def pupil_shift_um(phi_deg, R_rot_mm=DEFAULT_R_ROT_MM):
    phi_rad = np.radians(phi_deg)
    return R_rot_mm * 1000.0 * np.sin(phi_rad)
```

15.3 Ghost-separation formula

```
# Array_Design/array_geometry_lab/array_calculation/engine/
    ghost_formula.py
def delta_r_corrected_from_M_um(pitch_um, M_diopters, P0=60.0):
    return pitch_um * M_diopters / (P0 - M_diopters)
```

Validated to $R^2 = 0.995135$ against an independent two-hole coherent wave-optics propagation (32 measured points), against $R^2 = -1701.71$ for the disproven legacy formula (source: `ghost_formula.py` module docstring; see Chapter 14).

15.4 Axial myopia model

```
# stage2_fixed/myopia_axial.py
def axial_length_for_myopia(M_diopters, P0=60.0, n_prime=1.336):
    if M_diopters >= P0:
        raise ValueError(...)
    L0 = n_prime / P0
    L_ax = n_prime / (P0 - M_diopters)
    f_eye = L0
    return {"L_ax": L_ax, "f_eye": f_eye, ...}
```

15.5 The lens-phase / effective-wavelength pairing (root of Sections 10.1, 10.2)

```
# stage2_fixed/phase2_retina.py
def lens_phase(field, dx, wavelength, f_eye):
    """Thin-lens phase transmittance exp(-i k (x^2+y^2) / (2 f_eye)).
    `wavelength` here MUST be the image-space effective wavelength
    (wavelength/n_prime), never the raw vacuum wavelength. f_eye =
        n_prime/P0
    is already a vergence-equation distance measured through the
        vitreous --
    pairing it with the vacuum k under-applies the lens's converging
        phase by
    a factor of n_prime, shifting the true geometric focus to f_eye*
        n_prime
    instead of f_eye -- while propagate_to_retina still only propagates
        to
    f_eye at zero defocus, so the retina plane is evaluated short of
        the
    (wrongly shifted) focus. See PHYSICS_AUDIT.md item A for the full
    derivation and why the zero-defocus Airy validation gate, run at
        exact
    focus, did not surface it (item D)."""
    N = field.shape[0]
    L = N * dx
    X, Y, _, _ = junction.make_grid(N, L)
    k = 2 * np.pi / wavelength
    phase = np.exp(-1j * k * (X**2 + Y**2) / (2.0 * f_eye))
    return field * phase
```

This single docstring is the precise mechanism behind both Section 10.1 (a naive single-FFT implementation, evaluated at the wrong wavelength/plane pairing, would silently land exactly on the unshifted focus) and part of why Section 10.2's shadowing defect went undetected by this project's own zero-defocus ($M = 0$) Fraunhofer validation gate (Section 2.8): that gate is run *at* exact focus, where a wavelength-pairing error of this exact form produces no visible discrepancy – it only manifests at nonzero M, precisely the condition the validation gate does not test.

15.6 The ASM propagator itself (Equation 2.7), verbatim

```
# stage1_v2/physics.py
def asm_transfer_function(N, dx, wavelength, z, L, band_limit=True):
    """H(fx,fy) = exp(i k z sqrt(1 - (lambda fx)^2 - (lambda fy)^2)) on
    propagating orders, zero elsewhere (evanescent cutoff) -- the exact
    scalar ASM kernel, no paraxial term. When band_limit is True, also
    zeros propagating orders beyond the Matsushima & Shimobaba (2009)
    elliptical band limit f_limit = 1/(lambda*sqrt((2z/L)^2+1))."""
    FX, FY = _frequency_grid(N, dx)
    freq_sq = FX**2 + FY**2
    propagating = freq_sq <= (1.0 / wavelength**2)
    kz = np.zeros_like(freq_sq)
    kz[propagating] = np.sqrt(1.0 - (wavelength**2) * freq_sq[
       propagating])
    k = 2 * np.pi / wavelength
    kept = propagating
    band_suppressed = np.zeros_like(propagating)
    if band_limit:
        du = 1.0 / L
        fx_limit = 1.0 / (wavelength * np.sqrt((2*du*z)**2 + 1.0))
        within_band = (FX/fx_limit)**2 + (FY/fx_limit)**2 <= 1.0
        kept = propagating & within_band
        band_suppressed = propagating & ~within_band
    ...

def asm_propagate(U0, dx, wavelength, z, L=None, band_limit=True,
   precomputed=None):
    """Returns (U(x,y,z), band_info). band_info reports energy-weighted
    band-limit impact -- the residual check of Section~2.6's keybox."""
    N = U0.shape[0]
    if L is None:
        L = N * dx
    H, propagating, band_suppressed, fx_limit = asm_transfer_function(
        N, dx, wavelength, z, L, band_limit)
    U0_hat = np.fft.fft2(U0)
    propagating_power = np.sum(np.abs(U0_hat[propagating])**2)
    suppressed_power = np.sum(np.abs(U0_hat[band_suppressed])**2) if
       band_limit else 0.0
    band_info = {
        "band_limit_applied": band_limit,
        "fx_limit": fx_limit,
        "fraction_energy_suppressed": (suppressed_power/
           propagating_power)
                                       if propagating_power > 0 else
                                          0.0,
    }
    Uz = np.fft.ifft2(U0_hat * H)
    return Uz, band_info
```

This is the literal code behind Equation 2.7 and the energy-accounting keybox of Section 2.6: `fraction_energy_suppressed` is exactly the $f_{\text{suppressed}}$ that section's residual check compares against.

15.7 Visual Strehl (VSOTF), the CSF-weighted OTF-similarity integral

```
# stage4_human_vision/visual_strehl.py
```

```
def compute_vsotf(freq_cpd, otf_real, ref_freq_cpd, ref_otf_real,
    neural_weights):
    common_freq = freq_cpd
    ref_interp = np.interp(common_freq, ref_freq_cpd, ref_otf_real)
    num = np.trapz(otf_real * neural_weights, common_freq)
    den = np.trapz(ref_interp * neural_weights, common_freq)
    return {"vsotf": float(num / den) if abs(den) > 1e-12 else 0.0,
        ...}
```

The CSF-weighted ratio of the actual (aberrated/diffracted) OTF to the diffraction- limited reference OTF, both integrated against the same neural (CSF) weighting – the quantity entering Chapter 3's $\mathrm{VSOTF}^{-0.1}$ term.

15.8 Model A gate thresholds

```
# Array_Design/array_geometry_lab/array_evaluation/comparison_metric/
    gates.py
GHOST_FORMULA_GRID_DX_UM = 1.46484375  # confirmed: 6mm window / 4096
    grid
GHOST_DELTA_R_UNCERTAINTY_UM = 2.0 * GHOST_FORMULA_GRID_DX_UM
MARGIN_SAFETY_MULTIPLE = 2.0
VIABILITY_MIN_FRACTION_OF_HOLE_AREA = 0.20

def ghost_margin_threshold(df):
    return MARGIN_SAFETY_MULTIPLE * GHOST_DELTA_R_UNCERTAINTY_UM / df.
        resolvable_blur_um

def gate_ghost_safety(df):
    return df.worst_case_ghost_margin >= ghost_margin_threshold(df)

def gate_viability(df):
    return df.min_illuminated_area_um2_across_phi >=
        viability_area_threshold(df)
```

15.9 Ring-offset concentric: the alternating angular-offset construction

```
# Array_Design/array_geometry_lab/array_calculation/generators/
    ring_offset_concentric.py
def generate(N, a_hole_um, ring_spacing_um, holes_per_ring0):
    pts = [np.array([0.0, 0.0])]
    k = 1
    while len(pts) < N:
        n_k = max(1, int(round(holes_per_ring0 * k)))
        r_k = k * ring_spacing_um
        angular_step = 2 * np.pi / n_k
        offset = (angular_step / 2.0) if (k % 2 == 1) else 0.0
        thetas = np.linspace(0.0, 2 * np.pi, n_k, endpoint=False) +
            offset
        pts.extend([np.array([r_k*np.cos(t), r_k*np.sin(t)]) for t in
            thetas])
        k += 1
    ...
```

15.10 Concentric-centered: the recenter-before-prepend fix (Section 5.8.6)

```
# Array_Design/array_geometry_lab/array_calculation/generators/
    concentric_centered.py
# Ring points recentered on THEIR OWN centroid first, THEN the (0,0)
    centre hole is
# prepended, THEN finalize() runs once more on the combined set -- so
    the combined
# set's mean is exactly (0,0) by construction, and the final recenter
    step is a
# provable no-op. (Full ring-count/radius formula identical to
    ring_offset_concentric
# above; only the ordering of recenter-vs-prepend differs.)
```

15.11 Effective-density / Pareto binding-margin sentinel

```
# Array_Design/array_geometry_lab/array_evaluation/comparison_metric/
    metrics.py
def _binding_margin_with_sentinel(survivors):
    ghost = survivors.worst_case_ghost_margin.to_numpy(dtype=float)
    vign = survivors.worst_case_vignetting_margin.to_numpy(dtype=float)
    finite_vals = np.concatenate([ghost[np.isfinite(ghost)], vign[np.
        isfinite(vign)]])
    if len(finite_vals) == 0:
        sentinel = 1.0   # degenerate: every margin in the survivor set
            is +inf
    else:
        sentinel = 10.0 * finite_vals.max()
    ...
```

15.12 Model B: active-hole and blindness classification

```
# Array_Design/array_geometry_lab/array_calculation/
    compute_gaze_weighted_metrics.py
areas = vo.overlap_area_um2(delta, R_p, R_h)      # (n_samples, N_holes
    )
total_area = areas.sum(axis=1)
fractions = areas / hole_area
n_active = (fractions >= active_frac_threshold).sum(axis=1)

is_blind = (total_area == 0.0)
is_ghost = (n_active >= 2)
time_weighted_blind_pct = float(np.dot(BIN_WEIGHTS, frac_blind_per_bin)
    )
time_weighted_ghost_pct = float(np.dot(BIN_WEIGHTS, frac_ghost_per_bin)
    )
```

Chapter 16

Formula and Constants Quick-Reference

Consolidated from the derivations of Chapters 2–5, cross-checked against `Physics_Guide/MASTER_HANDBOOK.tex`'s own equivalent tables (Chapters 9–10 of that document, verified present, mtime 2026-09-11) and against live code where a discrepancy would matter (Chapter 14).

Table 16.1: Every working formula referenced in this paper, with status.

Quantity	Formula	Status
Myopic axial length	$L_{\text{ax}} = n'/(P_0 - M)$	live
Emmetropic axial length	$L_0 = n'/P_0 = 22.267\,\text{mm}$	live
ASM propagator (Eq. 2.7)	$H = \exp(ikz\sqrt{1 - \lambda^2(f_x^2 + f_y^2)})$	live, used for both propagation stages
Fresnel number	$N_F = R_{\text{hole}}^2/(\lambda z)$	live (regime gate)
Airy intensity	$I = I_0(2J_1(x)/x)^2$	validation only
Encircled energy	$EE = 1 - J_0^2(x) - J_1^2(x)$	validation only
On-axis Fresnel oscillation	$I(0,z) = 4I_0 \sin^2(\pi N_F/2)$	live (cross-check)
Circle–circle overlap (Eq. 5.7)	three-regime exact form	live
Gaze pupil shift (Eq. 5.6)	$\Delta = R_{\text{rot}} \sin\varphi$	live
Ghost separation	$\Delta r = sM/(P_0 - M)$	live, $R^2 = 0.995$
Acuity (recommended, Ch. 3)	$\text{MAR} = (f_{20/20}/f_{\text{cutoff},10\%})\,\text{VSOTF}^{-0.1}$	live
Brightness correction (Eq. 3.4)	$0.060 \log_{10}(300/E_{\text{td}})$ below knee	live, cited
Geometric blur (old)	$r_{\text{geo}} = \frac{a_{\text{stop}}}{2}\frac{\lvert v_r - v_f\rvert}{v_f}$	deprecated (Sec. 2.10)
RSS combination (Sec. 2.3)	$\sqrt{r_{\text{geo}}^2 + r_{\text{diff}}^2}$	deprecated, 64.8% mean error
Quadrature a_{eff}	$\sqrt{a_{\text{hole}}^2 + (2.44\lambda d_{\text{gap}}/a_{\text{hole}})^2}$	deprecated
Ghosting three-regime (Eq. 5.5)	$N_{\text{eff}} \le 1$ OR $\text{MTF}(f_0) < \text{CSF}^{-1}(f_0)$	live, harmonic-extended (Sec. 5.4.1)

Table 16.2: Constants used throughout this paper.

Symbol	Value	Meaning
P_0	60.0 D	Emmetropic eye power, held fixed for every M (axial myopia model)
n'	1.336	Vitreous refractive index
D_{pupil}	4000 μm	Default photopic pupil diameter
R_{rot}	13–14 mm (13.5 mm used)	Eye rotation radius
λ	550 nm	Default design wavelength (vacuum)
d_{gap}	12 mm	Spectacle-plane vertex distance

Table 16.2: Constants used throughout this paper.

Symbol	Value	Meaning
$c_{\min}$	55 μm	Shen (2026) fusion floor – inherited as a non-binding sanity check, not used as a live design constraint in this paper's own results
Scene luminance	10,000 cd/m^2	Bright-daylight default
Weber–Fechner knee	300 Td	Van Nes & Bouman (1967)
Brightness slope	0.060 LogMAR/decade	Eidbury et al. (2016)
N_{grid} (corrected)	4096	Post-Section 2.7-fix ASM grid resolution
Active-hole fraction threshold	0.5	`margins/active_pairs.py`
Ghost-margin safety multiple	2.0	`gates.py` (Section 6.1)
Viability floor	20% of one hole's own area	`gates.py`
Blindness gate tolerance (Model B)	1.0%	Section 7.3

Chapter 17

Full Acuity Sweep Table

Complete, unabridged data underlying every acuity result summarized elsewhere in this paper (Table 9.1, Figure 9.1), included in full per this paper's own provenance discipline.

Table 17.1: Full 120-point acuity sweep (15 apertures x 8 myopia powers), all fields, verbatim from `new_sweeps/acuity_sweep_stage4.csv` (mtime 2026-09-13 00:11).

a_{hole} (μm)	M (D)	LogMAR	MAR (arcmin)	VSOTF
75	0	1.1176	13.109	1.0000
100	0	0.9773	9.492	1.0000
150	0	0.7809	6.038	1.0000
200	0	0.6421	4.387	1.0000
300	0	0.4666	2.928	1.0000
400	0	0.3416	2.196	1.0000
500	0	0.2447	1.757	1.0000
600	0	0.1656	1.464	1.0000
700	0	0.0987	1.255	1.0000
750	0	0.0687	1.171	1.0000
1000	0	-0.0563	0.879	1.0000
1250	0	-0.1531	0.703	1.0000
1500	0	-0.2322	0.586	1.0000
1750	0	-0.2991	0.502	1.0000
2000	0	-0.3570	0.440	1.0000
75	1	1.1127	12.964	0.9919
100	1	0.9725	9.386	0.9924
150	1	0.7759	5.970	0.9932
200	1	0.6372	4.337	0.9938
300	1	0.4616	2.895	0.9940
400	1	0.3368	2.172	0.9925
500	1	0.2404	1.739	0.9885
600	1	0.1621	1.452	0.9816
700	1	0.0964	1.249	0.9710
750	1	0.0674	1.168	0.9641
1000	1	-0.0492	0.893	0.9121
1250	1	-0.1199	0.759	0.8312
1500	1	0.0758	1.191	0.7310
1750	1	0.2340	1.714	0.6279
2000	1	0.3364	2.170	0.5373
75	2	1.1077	12.815	0.9836
100	2	0.9675	9.279	0.9846
150	2	0.7710	5.902	0.9862
200	2	0.6322	4.288	0.9871
300	2	0.4568	2.863	0.9855
400	2	0.3327	2.151	0.9776
500	2	0.2377	1.729	0.9609
600	2	0.1620	1.452	0.9330

Table 17.1: Full 120-point acuity sweep (15 apertures x 8 myopia powers), all fields, verbatim from `new_sweeps/acuity_sweep_stage4.csv` (mtime 2026-09-13 00:11).

a_{hole} (μm)	M (D)	LogMAR	MAR (arcmin)	VSOTF
700	2	0.1011	1.262	0.8923
750	2	0.0756	1.190	0.8670
1000	2	0.1220	1.324	0.6998
1250	2	0.3941	2.478	0.5158
1500	2	0.5289	3.380	0.3843
1750	2	0.6276	4.243	0.3046
2000	2	0.7077	5.101	0.2444
75	3	1.1026	12.666	0.9752
100	3	0.9624	9.172	0.9767
150	3	0.7660	5.834	0.9789
200	3	0.6272	4.239	0.9797
300	3	0.4523	2.833	0.9743
400	3	0.3293	2.135	0.9553
500	3	0.2370	1.726	0.9174
600	3	0.1668	1.468	0.8571
700	3	0.1192	1.316	0.7744
750	3	0.1117	1.293	0.7261
1000	3	0.4710	2.958	0.4692
1250	3	0.6377	4.343	0.3058
1500	3	0.7481	5.598	0.2254
1750	3	0.8276	6.723	0.1674
2000	3	0.8937	7.829	0.1299
75	4	1.0975	12.518	0.9667
100	4	0.9573	9.064	0.9686
150	4	0.7609	5.766	0.9714
200	4	0.6222	4.190	0.9717
300	4	0.4479	2.805	0.9604
400	4	0.3268	2.122	0.9255
500	4	0.2388	1.733	0.8593
600	4	0.1808	1.516	0.7597
700	4	0.2728	1.874	0.6345
750	4	0.3916	2.464	0.5680
1000	4	0.6526	4.494	0.3051
1250	4	0.7894	6.158	0.2044
1500	4	0.8837	7.650	0.1397
1750	4	0.9593	9.106	0.1027
2000	4	1.0231	10.545	0.0783
75	5	1.0924	12.371	0.9580
100	5	0.9522	8.958	0.9604
150	5	0.7558	5.699	0.9635
200	5	0.6172	4.142	0.9630
300	5	0.4437	2.778	0.9437
400	5	0.3252	2.115	0.8885
500	5	0.2446	1.756	0.7887
600	5	0.2401	1.738	0.6490
700	5	0.4966	3.138	0.4929
750	5	0.5648	3.671	0.4201
1000	5	0.7767	5.980	0.2170
1250	5	0.8962	7.873	0.1381
1500	5	0.9863	9.689	0.0949
1750	5	1.0601	11.485	0.0686
2000	5	1.1224	13.255	0.0518
75	6	1.0872	12.224	0.9491
100	6	0.9469	8.850	0.9520
150	6	0.7506	5.631	0.9554
200	6	0.6122	4.094	0.9536
300	6	0.4399	2.754	0.9242
400	6	0.3250	2.113	0.8447

Table 17.1: Full 120-point acuity sweep (15 apertures x 8 myopia powers), all fields, verbatim from `new_sweeps/acuity_sweep_stage4.csv` (mtime 2026-09-13 00:11).

a_{hole} (μm)	M (D)	LogMAR	MAR (arcmin)	VSOTF
500	6	0.2585	1.814	0.7086
600	6	0.4667	2.929	0.5344
700	6	0.6297	4.263	0.3677
750	6	0.6876	4.871	0.3028
1000	6	0.8670	7.363	0.1657
1250	6	0.9812	9.576	0.0999
1500	6	1.0693	11.729	0.0676
1750	6	1.1417	13.859	0.0487
2000	6	1.2028	15.951	0.0366
75	8	1.0765	11.926	0.9310
100	8	0.9362	8.634	0.9348
150	8	0.7401	5.496	0.9383
200	8	0.6021	4.000	0.9326
300	8	0.4331	2.711	0.8764
400	8	0.3307	2.141	0.7394
500	8	0.4876	3.074	0.5342
600	8	0.6930	4.932	0.3299
700	8	0.8093	6.446	0.2053
750	8	0.8506	7.089	0.1764
1000	8	1.0048	10.112	0.0938
1250	8	1.1134	12.984	0.0577
1500	8	1.1983	15.787	0.0391
1750	8	1.2685	18.558	0.0279
2000	8	1.3300	21.379	0.0209

Chapter 18

Full Barten-CSF Ghosting Sweep Table

Complete, unabridged Pass 2 (corrected Barten CSF) data underlying the reclassification result of Section 5.4 (Table, Section 5.4: 16 Regime A, 0 Regime B, 20 Regime C). M=4D, d_{gap}=12mm, λ=550nm, D_{pupil}=4000μm, square packing, throughout.

Table 18.1: Full 36-point Barten-CSF ghosting-regime sweep (4 apertures x 9 pitches, M=4D, verbatim from `Array_Design/ghosting/ghosting_sweep_barten.csv`, mtime 2026-07-29 22:04). Regime per Equation 5.5.

a_{hole} (μm)	pitch (μm)	Regime	N_{eff} (worst)	f_0 (cpd)	MTF(f_0)	CSF thresh.
300	700	C	23	0.299	0.9652	0.0101
300	900	C	13	0.233	0.9747	0.0130
300	1200	C	8	0.175	0.9840	0.0172
300	1500	C	4	0.140	0.9894	0.0215
300	2000	C	2	0.105	0.9949	0.0286
300	2500	A	1	–	–	–
300	3200	A	0	–	–	–
300	4000	A	0	–	–	–
300	4800	A	0	–	–	–
400	700	C	23	0.299	0.9718	0.0101
400	900	C	13	0.233	0.9797	0.0130
400	1200	C	8	0.175	0.9872	0.0172
400	1500	C	4	0.140	0.9916	0.0215
400	2000	C	2	0.105	0.9959	0.0286
400	2500	A	1	–	–	–
400	3200	A	0	–	–	–
400	4000	A	0	–	–	–
400	4800	A	0	–	–	–
700	700	C	23	0.299	0.9754	0.0101
700	900	C	13	0.233	0.9832	0.0130
700	1200	C	8	0.175	0.9898	0.0172
700	1500	C	4	0.140	0.9933	0.0215
700	2000	C	2	0.105	0.9969	0.0286
700	2500	A	1	–	–	–
700	3200	A	0	–	–	–
700	4000	A	0	–	–	–
700	4800	A	0	–	–	–
1000	700	C	23	0.299	0.9684	0.0101
1000	900	C	13	0.233	0.9794	0.0130
1000	1200	C	8	0.175	0.9880	0.0172
1000	1500	C	4	0.140	0.9922	0.0215
1000	2000	C	2	0.105	0.9964	0.0286
1000	2500	A	1	–	–	–
1000	3200	A	0	–	–	–

Table 18.1: Full 36-point Barten-CSF ghosting-regime sweep (4 apertures x 9 pitches, M=4D, verbatim from `Array_Design/ghosting/ghosting_sweep_barten.csv`, mtime 2026-07-29 22:04). Regime per Equation 5.5.

a_{hole} (μm)	pitch (μm)	Regime	N_{eff} (worst)	f_0 (cpd)	MTF(f_0)	CSF thresh.
1000	4000	A	0	–	–	–
1000	4800	A	0	–	–	–

Chapter 19

Additional Full Results Tables

Three further live results files, referenced in passing in Chapters 6–7 but not shown in full there, reproduced here in full per this paper's provenance discipline.

19.1 Family ranking: best-of-eleven per myopia power

Source: `array_data/evaluation_results/family_ranking.csv` (1258 rows total, mtime 2026-09-13 00:37) – too large to reproduce in full; the rank-1 (highest packing density, density-only ranking, no safety gate applied) row per myopia power is shown below, computed directly from the live file.

M (D)	Rank-1 family	a_{hole} (μm)	Families competing
0	square	2000	10
1	square	1250	8
2	square	750	6
3	square	750	6
4	square	600	7
5	square	600	7
6	square	500	6
8	square	400	6
global	square	2000	8

Square wins the pure density-only ranking at every myopia power – consistent with Model A/Metric 1's own reported result (Section 6.3), confirming that result is not an artifact of Model A's own gating step, since this ranking applies no safety gate at all. The number of families with at least one candidate surviving to be ranked varies (6–10) across myopia powers, reflecting that not every family produces a viable candidate at every aperture/spacing combination tested.

19.2 Effective-density ranking, both fusion-threshold variants

Source: `effective_density_winner_per_M_cmin0.csv` and `..._cmin55.csv` (mtime 2026-09-13 01:01, both). Ranks by $n_{\text{active_holes},\varphi=0}/A_{\text{hull}}$ (only holes that actually admit light, not raw hole count) rather than raw packing density – the correction this ranking script was built to apply, after a confirmed finding that every prior margin-aware winner has exactly one active hole at $\varphi = 0$ (Section 2.11), making raw packing density count many structurally inert holes.

M (D)	Family ($c_{\min} = 0$)	Family ($c_{\min} = 55\,\mu$m)	Blind rate (both variants)
0	hexagonal	ring_offset_concentric	0.173 / 0.156
1	ring_offset_concentric	square	0.113 / 0.082
2	ring_offset_concentric	ring_offset_concentric	0.107
3	ring_offset_concentric	ring_offset_concentric	0.107
4	ring_offset_concentric	ring_offset_concentric	0.105
5	ring_offset_concentric	ring_offset_concentric	0.105
6	ring_offset_concentric	ring_offset_concentric	0.103
8	ring_offset_concentric	ring_offset_concentric	0.101
global	hexagonal	hexagonal	0.173

Notably different winners from both the pure-density ranking above and from Model B (Chapter 7): this metric rewards *active*-hole density specifically, and `ring_offset_concentric` – despite the single-axis blackout risk documented in Section 6.2 for a *different* specific candidate of this same family – wins most M values here. This is reported as a genuine, unresolved tension between evaluation criteria, not smoothed into a single "best" answer: Model B (Chapter 7) remains this project's primary standard specifically because it is the only one of these rankings that directly measures gaze-time-weighted safety rather than a density proxy for it.

19.3 Metric 2: the full density/margin Pareto front

Source: `metric2_pareto_front_per_M.csv` (18 rows total, mtime 2026-09-13 00:37) – the complete 2D (density, binding-margin) Pareto front per myopia power, shown here in full since it is small enough to reproduce entirely.

M (D)	Family	a_{hole} (μm)	Binding margin
0	square, rectangular (tied)	2000	1.0
1	square, rectangular (tied)	1250	1.0
2	square, rectangular (tied)	750	1.0
3	square, rectangular (tied)	750	1.0
4	square, rectangular (tied)	600	1.0
5	square, rectangular (tied)	600	1.0
6	square, rectangular (tied)	500	1.0
8	square, rectangular (tied)	400	1.0
global	square, rectangular (tied)	2000	1.0

Every Pareto front in this project's own data has exactly 2 points (never more), and they are square and rectangular *tied* at every myopia power – direct numerical confirmation of the exact-equivalence identity these two generators are built to satisfy (Section 5.8: rectangular reduces to square exactly at equal x/y pitch), not a coincidence of the ranking.

`binding_margin` is defined as min(worst_case_ghost_margin, worst_case_vignetting_margin) per candidate (`metrics.py`, `_binding_margin_with_sentinel`, verified read directly) – but both underlying margins are $+\infty$ for every row shown above (structurally safe, zero active hole pairs, Section 6.3). Since a finite sentinel value is needed for these rows to participate in ordinary 2D Pareto-dominance comparison (letting raw $+\infty$ into the comparison would make every such row trivially co-dominant regardless of density, collapsing the density/margin tradeoff this metric exists to show), and *every* candidate in this survivor set has an infinite margin, there is no finite margin value anywhere in the set to scale a sentinel from – the implementation's own documented fallback for exactly this degenerate case is a fixed sentinel of 1.0, used here because ranking then falls entirely to density (the two tied candidates' shared density value, not shown in this abbreviated table but present in the source CSV). The uniform 1.0 seen above is this documented fallback behaviour, not a tautology of the underlying formula.

Chapter 20

Full Ghosting Sweep Tables: Passes 1 and 3

Pass 2 (the corrected Barten-CSF sweep) is given in full in Chapter 18. The other two passes of the same investigation (Section 5.4–5.4.1) are reproduced here in full for the same reason.

20.1 Pass 1: the heuristic-CSF sweep that motivated the investigation

Table 20.1: Full 36-point Pass 1 (heuristic, uncorrected CSF) sweep – verbatim from `Array_Design/ghosting/ghosting_sweep.csv` (mtime 2026-07-29 21:53). Zero Regime C found; root cause in Section 5.4.

a_{hole} (μm)	pitch (μm)	Regime	N_{eff} (worst)
300	700	B	23
300	900	B	13
300	1200	B	8
300	1500	B	4
300	2000	B	2
300	2500	A	1
300	3200	A	0
300	4000	A	0
300	4800	A	0
400	700	B	23
400	900	B	13
400	1200	B	8
400	1500	B	4
400	2000	B	2
400	2500	A	1
400	3200	A	0
400	4000	A	0
400	4800	A	0
700	700	B	23
700	900	B	13
700	1200	B	8
700	1500	B	4
700	2000	B	2
700	2500	A	1
700	3200	A	0
700	4000	A	0
700	4800	A	0
1000	700	B	23
1000	900	B	13
1000	1200	B	8

Table 20.1: Full 36-point Pass 1 (heuristic, uncorrected CSF) sweep – verbatim from `Array_Design/ghosting/ghosting_sweep.csv` (mtime 2026-07-29 21:53). Zero Regime C found; root cause in Section 5.4.

a_{hole} (μm)	pitch (μm)	Regime	N_{eff} (worst)
1000	1500	B	4
1000	2000	B	2
1000	2500	A	1
1000	3200	A	0
1000	4000	A	0
1000	4800	A	0

20.2 Pass 3: the harmonic-extended sweep

Table 20.2: Full 36-point Pass 3 (harmonic-extended) sweep – verbatim from `Array_Design/ghosting/ghosting_sweep_harmonic.csv` (mtime 2026-07-29 22:11). Confirms the worst-case harmonic is never $n = 1$ (Section 5.4.1).

a_{hole}	pitch	Regime	N_{eff}	worst n	worst margin	fund. margin
300	700	C	23	7.0000	329.2000	95.3000
300	900	C	13	9.0000	329.2000	75.2000
300	1200	C	8	12.0000	329.2000	57.1000
300	1500	C	4	15.0000	329.2000	46.0000
300	2000	C	2	20.0000	329.2000	34.7000
300	2500	A	1	–	–	–
300	3200	A	0	–	–	–
300	4000	A	0	–	–	–
300	4800	A	0	–	–	–
400	700	C	23	7.0000	329.2000	95.9000
400	900	C	13	9.0000	329.2000	75.6000
400	1200	C	8	12.0000	329.2000	57.3000
400	1500	C	4	14.0000	329.4000	46.1000
400	2000	C	2	19.0000	329.6000	34.8000
400	2500	A	1	–	–	–
400	3200	A	0	–	–	–
400	4000	A	0	–	–	–
400	4800	A	0	–	–	–
700	700	C	23	5.0000	273.0000	96.3000
700	900	C	13	6.0000	272.9000	75.8000
700	1200	C	8	8.0000	272.9000	57.4000
700	1500	C	4	10.0000	272.9000	46.2000
700	2000	C	2	14.0000	273.2000	34.8000
700	2500	A	1	–	–	–
700	3200	A	0	–	–	–
700	4000	A	0	–	–	–
700	4800	A	0	–	–	–
1000	700	C	23	4.0000	218.4000	95.6000
1000	900	C	13	5.0000	219.3000	75.5000
1000	1200	C	8	6.0000	218.5000	57.3000
1000	1500	C	4	8.0000	219.6000	46.1000
1000	2000	C	2	11.0000	219.5000	34.8000
1000	2500	A	1	–	–	–
1000	3200	A	0	–	–	–
1000	4000	A	0	–	–	–
1000	4800	A	0	–	–	–

Chapter 21

Full Fabrication Hole-Coordinate Tables

Every hole center coordinate actually cut into every fabricated mask, verbatim from `Array_Design/array_geometry_lab/fabricatio` (all 9 files, mtime 2026-09-13 00:58, re-verified before this dump). Target etched diameter and the undercut-compensated mask-aperture bracket (Section 8.1.1) are given per row; these are identical within one file since every hole in one design shares one aperture size (Section 2.11), but are reproduced per-row here rather than factored out, matching the source files own layout exactly.

21.1 $M = 0$D (20 holes)

Table 21.1: Hole coordinates, $M = 0$D, from `hole_coordinates_M0.0.csv`.

idx	x (mm)	y (mm)	target diam. (μm)
0	0.40000	0.20000	2000
1	0.40000	4.20000	2000
2	4.40000	0.20000	2000
3	-3.60000	0.20000	2000
4	0.40000	-3.80000	2000
5	4.40000	4.20000	2000
6	-3.60000	4.20000	2000
7	4.40000	-3.80000	2000
8	-3.60000	-3.80000	2000
9	0.40000	8.20000	2000
10	8.40000	0.20000	2000
11	0.40000	-7.80000	2000
12	-7.60000	0.20000	2000
13	-3.60000	-7.80000	2000
14	-7.60000	4.20000	2000
15	-3.60000	8.20000	2000
16	4.40000	8.20000	2000
17	-7.60000	-3.80000	2000
18	8.40000	-3.80000	2000
19	4.40000	-7.80000	2000

21.2 $M = 1$D (20 holes)

Table 21.2: Hole coordinates, M = 1D, from `hole_coordinates_M1.0.csv`.

idx	x (mm)	y (mm)	target diam. (μm)
0	-0.10000	0.51962	1250
1	1.90000	3.98372	1250
2	-2.10000	3.98372	1250
3	1.90000	-2.94449	1250
4	-2.10000	-2.94449	1250
5	3.90000	0.51962	1250
6	-4.10000	0.51962	1250
7	-0.10000	7.44782	1250
8	-0.10000	-6.40859	1250
9	-6.10000	-2.94449	1250
10	5.90000	3.98372	1250
11	-6.10000	3.98372	1250
12	5.90000	-2.94449	1250
13	3.90000	-6.40859	1250
14	-4.10000	-6.40859	1250
15	3.90000	7.44782	1250
16	-4.10000	7.44782	1250
17	7.90000	0.51962	1250
18	-8.10000	0.51962	1250
19	1.90000	-9.87269	1250

21.3 $M = 2$D (20 holes)

Table 21.3: Hole coordinates, M = 2D, from `hole_coordinates_M2.0.csv`.

idx	x (mm)	y (mm)	target diam. (μm)
0	-0.10000	0.51962	750
1	1.90000	3.98372	750
2	-2.10000	3.98372	750
3	1.90000	-2.94449	750
4	-2.10000	-2.94449	750
5	3.90000	0.51962	750
6	-4.10000	0.51962	750
7	-0.10000	7.44782	750
8	-0.10000	-6.40859	750
9	-6.10000	-2.94449	750
10	5.90000	3.98372	750
11	-6.10000	3.98372	750
12	5.90000	-2.94449	750
13	3.90000	-6.40859	750
14	-4.10000	-6.40859	750
15	3.90000	7.44782	750
16	-4.10000	7.44782	750
17	7.90000	0.51962	750
18	-8.10000	0.51962	750
19	1.90000	-9.87269	750

21.4 $M = 3$D (20 holes)

Table 21.4: Hole coordinates, M = 3D, from `hole_coordinates_M3.0.csv`.

idx	x (mm)	y (mm)	target diam. (μm)
0	-0.10000	0.51962	750
1	1.90000	3.98372	750
2	-2.10000	3.98372	750
3	1.90000	-2.94449	750
4	-2.10000	-2.94449	750
5	3.90000	0.51962	750
6	-4.10000	0.51962	750
7	-0.10000	7.44782	750
8	-0.10000	-6.40859	750
9	-6.10000	-2.94449	750
10	5.90000	3.98372	750
11	-6.10000	3.98372	750
12	5.90000	-2.94449	750
13	3.90000	-6.40859	750
14	-4.10000	-6.40859	750
15	3.90000	7.44782	750
16	-4.10000	7.44782	750
17	7.90000	0.51962	750
18	-8.10000	0.51962	750
19	1.90000	-9.87269	750

21.5 $M = 4$D (20 holes)

Table 21.5: Hole coordinates, M = 4D, from `hole_coordinates_M4.0.csv`.

idx	x (mm)	y (mm)	target diam. (μm)
0	-0.10000	0.51962	600
1	1.90000	3.98372	600
2	-2.10000	3.98372	600
3	1.90000	-2.94449	600
4	-2.10000	-2.94449	600
5	3.90000	0.51962	600
6	-4.10000	0.51962	600
7	-0.10000	7.44782	600
8	-0.10000	-6.40859	600
9	-6.10000	-2.94449	600
10	5.90000	3.98372	600
11	-6.10000	3.98372	600
12	5.90000	-2.94449	600
13	3.90000	-6.40859	600
14	-4.10000	-6.40859	600
15	3.90000	7.44782	600
16	-4.10000	7.44782	600
17	7.90000	0.51962	600
18	-8.10000	0.51962	600
19	1.90000	-9.87269	600

21.6 $M = 5$D (20 holes)

Table 21.6: Hole coordinates, M = 5D, from `hole_coordinates_M5.0.csv`.

idx	x (mm)	y (mm)	target diam. (μm)
0	-0.10000	0.51962	600
1	1.90000	3.98372	600
2	-2.10000	3.98372	600
3	1.90000	-2.94449	600
4	-2.10000	-2.94449	600
5	3.90000	0.51962	600
6	-4.10000	0.51962	600
7	-0.10000	7.44782	600
8	-0.10000	-6.40859	600
9	-6.10000	-2.94449	600
10	5.90000	3.98372	600
11	-6.10000	3.98372	600
12	5.90000	-2.94449	600
13	3.90000	-6.40859	600
14	-4.10000	-6.40859	600
15	3.90000	7.44782	600
16	-4.10000	7.44782	600
17	7.90000	0.51962	600
18	-8.10000	0.51962	600
19	1.90000	-9.87269	600

21.7 $M = 6$D (100 holes)

Table 21.7: Hole coordinates, M = 6D, from `hole_coordinates_M6.0.csv`.

idx	x (mm)	y (mm)	target diam. (μm)
0	0.04000	0.20785	500
1	-1.96000	3.67195	500
2	2.04000	3.67195	500
3	2.04000	-3.25626	500
4	-1.96000	-3.25626	500
5	-3.96000	0.20785	500
6	4.04000	0.20785	500
7	0.04000	7.13605	500
8	0.04000	-6.72036	500
9	-5.96000	-3.25626	500
10	6.04000	-3.25626	500
11	6.04000	3.67195	500
12	-5.96000	3.67195	500
13	4.04000	-6.72036	500
14	-3.96000	-6.72036	500
15	4.04000	7.13605	500
16	-3.96000	7.13605	500
17	-7.96000	0.20785	500
18	8.04000	0.20785	500
19	2.04000	-10.18446	500
20	-1.96000	-10.18446	500
21	-1.96000	10.60015	500
22	2.04000	10.60015	500
23	8.04000	-6.72036	500
24	-9.96000	3.67195	500
25	10.04000	-3.25626	500
26	10.04000	3.67195	500
27	-7.96000	7.13605	500
28	8.04000	7.13605	500

Table 21.7: Hole coordinates, M = 6D, from `hole_coordinates_M6.0.csv`.

idx	x (mm)	y (mm)	target diam. (μm)
29	-9.96000	-3.25626	500
30	-7.96000	-6.72036	500
31	6.04000	-10.18446	500
32	-5.96000	-10.18446	500
33	-5.96000	10.60015	500
34	6.04000	10.60015	500
35	12.04000	0.20785	500
36	-11.96000	0.20785	500
37	0.04000	14.06425	500
38	0.04000	-13.64856	500
39	12.04000	7.13605	500
40	-11.96000	7.13605	500
41	-11.96000	-6.72036	500
42	12.04000	-6.72036	500
43	-3.96000	-13.64856	500
44	10.04000	-10.18446	500
45	-3.96000	14.06425	500
46	10.04000	10.60015	500
47	4.04000	-13.64856	500
48	-9.96000	10.60015	500
49	-9.96000	-10.18446	500
50	4.04000	14.06425	500
51	14.04000	-3.25626	500
52	-13.96000	-3.25626	500
53	-13.96000	3.67195	500
54	14.04000	3.67195	500
55	8.04000	14.06425	500
56	-7.96000	14.06425	500
57	8.04000	-13.64856	500
58	-7.96000	-13.64856	500
59	16.04000	0.20785	500
60	-15.96000	0.20785	500
61	-13.96000	-10.18446	500
62	16.04000	-6.72036	500
63	-13.96000	10.60015	500
64	-15.96000	-6.72036	500
65	-1.96000	-17.11266	500
66	14.04000	-10.18446	500
67	16.04000	7.13605	500
68	2.04000	-17.11266	500
69	-15.96000	7.13605	500
70	14.04000	10.60015	500
71	-1.96000	17.52835	500
72	2.04000	17.52835	500
73	-17.96000	-3.25626	500
74	-5.96000	17.52835	500
75	18.04000	3.67195	500
76	-17.96000	3.67195	500
77	6.04000	17.52835	500
78	12.04000	14.06425	500
79	-11.96000	14.06425	500
80	-5.96000	-17.11266	500
81	18.04000	-3.25626	500
82	12.04000	-13.64856	500
83	6.04000	-17.11266	500
84	-11.96000	-13.64856	500
85	-9.96000	-17.11266	500
86	10.04000	17.52835	500
87	10.04000	-17.11266	500

Table 21.7: Hole coordinates, M = 6D, from `hole_coordinates_M6.0.csv`.

idx	x (mm)	y (mm)	target diam. (μm)
88	-9.96000	17.52835	500
89	20.04000	0.20785	500
90	-19.96000	0.20785	500
91	0.04000	-20.57676	500
92	0.04000	20.99246	500
93	18.04000	-10.18446	500
94	18.04000	10.60015	500
95	-17.96000	-10.18446	500
96	-17.96000	10.60015	500
97	-3.96000	20.99246	500
98	-3.96000	-20.57676	500
99	4.04000	-20.57676	500

21.8 $M = 8$D (50 holes)

Table 21.8: Hole coordinates, M = 8D, from `hole_coordinates_M8.0.csv`.

idx	x (mm)	y (mm)	target diam. (μm)
0	-0.20000	0.20785	400
1	-2.20000	3.67195	400
2	1.80000	3.67195	400
3	1.80000	-3.25626	400
4	-2.20000	-3.25626	400
5	-4.20000	0.20785	400
6	3.80000	0.20785	400
7	-0.20000	7.13605	400
8	-0.20000	-6.72036	400
9	-6.20000	-3.25626	400
10	5.80000	-3.25626	400
11	5.80000	3.67195	400
12	-6.20000	3.67195	400
13	3.80000	-6.72036	400
14	-4.20000	-6.72036	400
15	3.80000	7.13605	400
16	-4.20000	7.13605	400
17	-8.20000	0.20785	400
18	7.80000	0.20785	400
19	1.80000	-10.18446	400
20	-2.20000	-10.18446	400
21	-2.20000	10.60015	400
22	1.80000	10.60015	400
23	7.80000	-6.72036	400
24	-10.20000	3.67195	400
25	9.80000	-3.25626	400
26	9.80000	3.67195	400
27	-8.20000	7.13605	400
28	7.80000	7.13605	400
29	-10.20000	-3.25626	400
30	-8.20000	-6.72036	400
31	5.80000	-10.18446	400
32	-6.20000	-10.18446	400
33	-6.20000	10.60015	400
34	5.80000	10.60015	400
35	11.80000	0.20785	400
36	-12.20000	0.20785	400

Table 21.8: Hole coordinates, M = 8D, from `hole_coordinates_M8.0.csv`.

idx	x (mm)	y (mm)	target diam. (μm)
37	-0.20000	-13.64856	400
38	-0.20000	14.06425	400
39	11.80000	-6.72036	400
40	-12.20000	7.13605	400
41	11.80000	7.13605	400
42	-12.20000	-6.72036	400
43	3.80000	-13.64856	400
44	-10.20000	-10.18446	400
45	9.80000	-10.18446	400
46	3.80000	14.06425	400
47	-4.20000	14.06425	400
48	9.80000	10.60015	400
49	-4.20000	-13.64856	400

21.9 $M = global$D (20 holes)

Table 21.9: Hole coordinates, M = *global*D, from `hole_coordinates_Mglobal.csv`.

idx	x (mm)	y (mm)	target diam. (μm)
0	0.40000	0.20000	2000
1	0.40000	4.20000	2000
2	4.40000	0.20000	2000
3	-3.60000	0.20000	2000
4	0.40000	-3.80000	2000
5	4.40000	4.20000	2000
6	-3.60000	4.20000	2000
7	4.40000	-3.80000	2000
8	-3.60000	-3.80000	2000
9	0.40000	8.20000	2000
10	8.40000	0.20000	2000
11	0.40000	-7.80000	2000
12	-7.60000	0.20000	2000
13	-3.60000	-7.80000	2000
14	-7.60000	4.20000	2000
15	-3.60000	8.20000	2000
16	4.40000	8.20000	2000
17	-7.60000	-3.80000	2000
18	8.40000	-3.80000	2000
19	4.40000	-7.80000	2000

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Kyoji Matsushima and Tomoyoshi Shimobaba. [needs verification: exact title/journal not recorded in the citing source – found only as an inline code-comment attribution "matsushima and shimobaba (2009)" in stage1_v2/physics.py's asm_transfer_function(), describing the elliptical band-limit formula used for anti-aliasing in this project's asm propagation (section 2.6). likely refers to their band-limited asm work published in optics express (2009), not independently re-verified against the primary source for this paper.], 2009. Source of the Matsushima band-limit formula, Section 2.6 of this paper.

N. Samee et al. Assessment of pinhole for visual acuity recoverment in patients with myopia, 2022. REJECTED as a data source after reading the primary text directly: a widely-repeated web-search-summary claim attributed to this paper ("0.17 LogMAR at up to 5.5D, 1.16 LogMAR at 7-11D, 1.2mm aperture") does NOT appear anywhere in the actual paper, which tests no aperture sweep and reports no LogMAR values at all. Documented here, and in stage4_human_vision/literature.py, specifically so this fabricated claim is not silently re-discovered and re-trusted.

[initial NEEDS VERIFICATION] Shen. [needs verification: exact paper title not found in repository – only author surname "shen", affiliation "hangzhou normal university", and date "aug. 2026" are recorded, in array_design/array_geometry_lab/shen_model/shen_evaluator.py line 3]. ResearchGate preprint (exact URL/DOI [NEEDS VERIFICATION]), 2026. Source of the RSS aperture-optimization approximation and the fusion threshold $c_{\min} = 55\,\mu$m used as a non-binding sanity check in this project (Section 7).

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